

Smith-Magenis syndrome protein *Rai1* regulates body weight homeostasis through hypothalamic *Bdnf*-producing neurons and TrkB signalling

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

About eLife's process

Reviewed preprint posted

August 23, 2023 (this version)

Posted to bioRxiv

July 31, 2023

Sent for peer review

July 6, 2023

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Abstract

Retinoic acid-induced 1 (RAI1) haploinsufficiency causes Smith-Magenis syndrome (SMS), a genetic disorder with symptoms including hyperphagia, hyperlipidemia, severe obesity, and autism phenotypes. *Rai1* is a transcriptional regulator with a pan-neural expression pattern and hundreds of downstream targets. The mechanisms linking neural *Rai1* to body weight regulation remain unclear. Here we find that hypothalamic brain-derived neurotrophic factor (*Bdnf*)-TrkB signalling is disrupted in SMS (*Rai1*^{+/-}) mice. Selective *Rai1* loss from all *Bdnf*-producing cells or from *Bdnf*-producing neurons in the paraventricular nucleus of the hypothalamus (PVH) induced obesity in mice. Electrophysiological recordings revealed that *Rai1* ablation increased inhibitory synaptic transmission to PVH^{Bdnf} neurons and decreased intrinsic neuronal excitability. Chronic treatment of SMS mice with a partial agonist of tropomyosin receptor kinase B (TrkB), the cognate *Bdnf* receptor, delayed obesity onset. This treatment also partially rescued disrupted lipid profiles, insulin intolerance, and stereotypical repetitive behaviour in SMS mice. These data argue that *Rai1* regulates body weight and metabolic function through hypothalamic *Bdnf*-producing neurons and that targeting TrkB signalling might improve associated SMS phenotypes.

eLife assessment

This **valuable** study informs whether diminishing BDNF expression or alterations in the activity of BDNF-containing neurons in the paraventricular nucleus of the hypothalamus contributes to metabolic alterations in individuals with reduced RAI1 function, including those afflicted with Smith-Magenis syndrome (SMS). The evidence supporting the conclusions is **compelling** in that RAI1 deficits in BDNF-containing neurons partly contribute, with prominent effects on glycemic control and modest effects on feeding and body weight regulation, however, the histological analyses of BDNF and TrkB expressions are **inadequate**. This study would be of interest to neuroscientists and medical biologists working on metabolic disorders such as obesity and diabetes, as the findings in this study further links SMS-associated obesity with reduced *Bdnf* gene expression in the PVH and shed light on the role of the *Rai1* gene in the PVH *Bdnf* neurons and offer a basis for future therapeutic strategies for managing obesity in SMS.

Introduction

Obesity is an increasing global concern: 20% of the world's adult population is expected to become obese by 2025 if current trends persist (Collaboration, 2017). However, the complex environmental, behavioural, and genetic contributions to obesity make this a difficult condition to model and address. Monogenic obesity disorders provide a consistent system where the pathophysiology, genetic predisposition, and neurobiology underlying phenotypes can potentially provide insights that translate to polygenic obesity (Loos & Yeo, 2022).

One such monogenic disorder is Smith-Magenis syndrome (SMS; OMIM #182290) (Smith *et al.*, 1986), a neurogenetic condition frequently associated with obesity. 80% of SMS cases are caused by interstitial deletions of the 17p11.2 chromosomal region containing a dosage-sensitive gene, *retinoic acid-induced 1 (RAI1)* (Slager *et al.*, 2003). More than 90% of SMS patients are overweight (40% > 95th percentile weight, 40% > 85th percentile weight) and exhibit over-eating tendencies by adolescence (Alaimo *et al.*, 2015; Burns *et al.*, 2010; Gandhi *et al.*, 2022). 20% of SMS patients do not have 17p11.2 deletions but instead carry *RAI1* point mutations that are still frequently associated with obesity (Boot *et al.*, 2021; Edelman *et al.*, 2007). These clinical data indicate that RAI is critical for energy homeostasis underlying body weight regulation.

The human RAI1 protein shares >80% sequence identity with the mouse *Rai1* protein, allowing us to study *Rai1* function *in vivo* (Huang *et al.*, 2016; Javed *et al.*, 2020). We previously found that *Rai1* encodes a transcriptional regulator that binds to the promoter or enhancer regions of its target genes, most of which are involved in neurodevelopment and synaptic transmission (Huang *et al.*, 2016). SMS (*Rai1*^{+/−}) mice exhibit obesity associated with hyperphagia and increased adiposity (Burns *et al.*, 2010; Huang *et al.*, 2016). Interestingly, *Rai1* may have a sexually dimorphic function, as female SMS mice exhibit more pronounced obesity than males (Burns *et al.*, 2010; Huang *et al.*, 2016).

Using a homozygous *Rai1* conditional knockout (*Rai1*-cKO) mice, we found that *Rai1* deletion in either subcortical excitatory neurons (vGlut2⁺) or single-minded homolog 1 (Sim1⁺) neurons induced obesity in mice (Huang *et al.*, 2016). Sim1⁺ neurons reside in several brain regions including the nucleus of the lateral olfactory tract, the supraoptic nucleus, the medial amygdala, and the paraventricular nucleus of the hypothalamus (PVH), a region that is critical for body weight regulation (Balthasar *et al.*, 2005). Sim1⁺ neurons in the PVH are composed of several

molecularly defined cell types including those that express oxytocin, vasopressin, corticotropin-releasing hormone, somatostatin, melanocortin-4-receptor (MC4R), growth hormone-releasing hormone (GHRH), or brain-derived neurotrophic factor (Bdnf) (An *et al.*, 2015 [↗](#)). Deleting *Bdnf* from the PVH results in hyperphagic obesity and reduced locomotion (An *et al.*, 2015 [↗](#)). While PVH neuronal activity is reduced during feeding (Xu *et al.*, 2019 [↗](#)), activation of PVH^{Bdnf} neurons results in weight loss (Wu & Xu, 2022 [↗](#)). We recently found that *Rai1* promotes *Bdnf* expression in hypothalamus throughout life (Javed *et al.*, 2021 [↗](#)). We hypothesized that *Rai1* may be connected to Bdnf-producing neurons in PVH, explaining how a neural transcription factor regulates body weight homeostasis.

Bdnf regulates energy metabolism by binding to its cognate receptor tropomyosin receptor kinase B (TrkB) (An *et al.*, 2015 [↗](#)) and may thus provide a potential therapeutic target for SMS. However, the link between *Rai1*, Bdnf, and SMS symptoms is currently unclear. Patients carrying monogenic mutations in either *BDNF* (Gray *et al.*, 2006 [↗](#)) or *NTRK2* (encoding TrkB) (Yeo *et al.*, 2004 [↗](#)) experience severe hyperphagic obesity. Moreover, obesity and associated metabolic dysfunction such as impaired glucose and insulin sensitivity are correlated with impaired PI3K-Akt pathway, a TrkB downstream target (Su *et al.*, 2021 [↗](#)). Given that (1) *Bdnf* overexpression during early embryonic development or (2) targeting *Bdnf* overexpression to the PVH of adolescent SMS (*Rai1*^{+/-}) mice protected them from weight gain and metabolic defects (Javed *et al.*, 2021 [↗](#)), and that (3) a point mutation in *RAI1* disrupts its ability to transcriptionally enhance *Bdnf* expression *in vitro* (Abad *et al.*, 2018 [↗](#)), pharmacological targeting of Bdnf signaling is an attractive and yet unexplored therapeutic avenue for SMS.

Building on our previous discoveries, here we examine the neurobiological functions of *Rai1* in Bdnf-producing neurons, specifically those that reside in the PVH. Importantly, we evaluated the involvement of the Bdnf-TrkB pathways in SMS pathogenesis using a small molecule drug (LM22A-4) to enhance TrkB signalling. Our work indicates that PVH^{Bdnf} neurons depend on *Rai1* to maintain normal neuronal activity and regulate body weight homeostasis, and TrkB signalling dysfunction contributes to obesity associated with SMS mice.

Results

***Rai1* deficiency dysregulates specific downstream targets of neurotrophin in the mouse hypothalamus**

Multiple lines of evidence indicated that SMS mice show decreased *Bdnf* mRNA expression in the hypothalamus (Burns *et al.*, 2010 [↗](#); Huang *et al.*, 2016 [↗](#); Javed *et al.*, 2021 [↗](#)). To determine if Bdnf protein levels were consequently altered in the hypothalamus, we performed enzyme-linked immunosorbent assay (ELISA) and found that Bdnf protein levels were significantly reduced in 7-week-old pre-symptomatic SMS mice relative to controls (Fig 1A [↗](#)). To quantify if molecular pathways associated with Bdnf-TrkB signalling were altered, we subjected hypothalamic tissue lysates from 7-week-old pre-obese SMS and control (Ctrl) mice to reverse phase protein array (RPPA) (Fig 1B [↗](#)). We found that control and SMS samples clustered separately in principle component analysis (Fig 1C [↗](#)).

Hierarchical clustering analysis indicated that the majority of differentially expressed phosphorylated proteins (Fig 1D [↗](#)) and total proteins (Figure 1—figure supplement 1 [↗](#)) were downregulated in SMS mice. We mapped all proteins, as well as phospho-proteins, that were differentially expressed in Ctrl and SMS samples onto the interconnected biological network and identified a subset of connected nodes downregulated in SMS (FDR < 10e-5) belong to the neurotrophin, PI3K-Akt, and mTOR signalling pathways (Fig 1E [↗](#)). These pathways are known to play a critical role in regulating body weight homeostasis and can be activated by Bdnf either directly (neurotrophin pathway) or indirectly (PI3K-Akt and mTOR pathways) (Cota *et al.*, 2006 [↗](#);

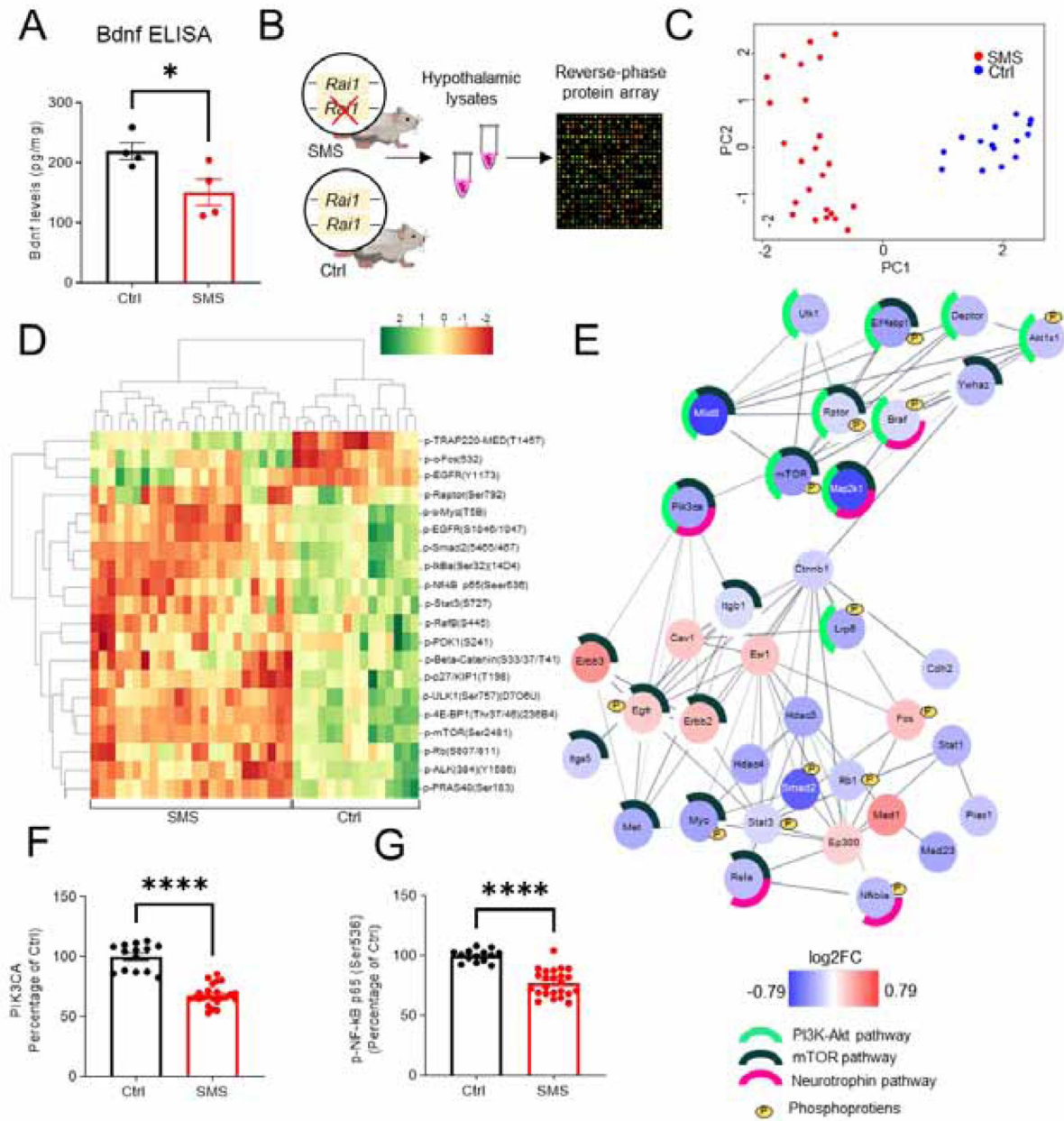


Figure 1

Rai1 haploinsufficiency disrupts hypothalamic proteomic profile in mice

(A) ELISA showing that Bdnf protein levels in hypothalamic tissues were significantly reduced in SMS mice compared to controls (n=4/genotype).

*p<0.05, student's t-test.

(B) A schematic diagram showing the experimental strategy. Hypothalamic tissue lysates from 7-week-old control (n = 5, 3 technical triplicates/sample) and SMS (n = 8, 3 technical triplicates/sample) mice were subjected to RPPA that utilizes a cocktail of 240 validated antibodies to probe several key intracellular signalling pathways by quantifying the expression levels of total and phosphorylated proteins

(C) Principal component analysis segregates the SMS and Ctrl groups where PC1 explains 47% of the variance, and PC2 explains 25% of the variance.

(D) Heat map showing hierarchical clustering of differentially expressed phospho-proteins: most showed reduced levels in SMS samples.

(E) Protein-protein interaction (PPI) analysis showing differentially expressed proteins and phospho-proteins in a crosstalk network. PPI enrichment p-value: < 1.0e-16. The nodes represent the proteins/phospho-proteins and the lines indicate physical interactions. Associations with specific molecular pathways are indicated as color coded outer rings.

(F-G) Protein levels of two neurotrophin pathway-related nodes, PIK3CA **(F)** and phospho-NF-kB **(G)**, which were significantly reduced in SMS samples.

****p<0.0001, student's t-test.

Schultze *et al.*, 2012 [\[1\]](#)). The expression levels of PIK3CA, a common node in these pathways, were significantly reduced in the SMS mice compared to controls (**Fig 1F** [\[1\]](#)). Moreover, the expression levels of phosphorylated NF- κ B (nuclear factor-kappa B), a downstream target of the neurotrophin pathway, were significantly reduced in the SMS group (**Fig 1G** [\[1\]](#)). Together, these results suggest that *Rai1* loss impairs specific pathways, including neurotrophin signalling, and indicate that dysfunction of Bdnf-producing cells could contribute to SMS pathogenesis.

Bdnf-producing cells depend on *Rai1* to regulate body weight homeostasis

To target Bdnf-producing cells, we used a *Bdnf*^{Cre/+} mouse line that carries a viral 2A oligopeptide and Cre recombinase immediately upstream of Bdnf's endogenous STOP codon (Tan *et al.*, 2016 [\[2\]](#)). We crossed the *Bdnf*^{Cre/+} mice with tdTomato-Ai9 reporter mice and found that Ai9 signals faithfully reflected endogenous Bdnf protein expression in the PVH (**Figure 2—figure supplement 1A-E** [\[1\]](#)). More than 50% of Bdnf-producing cells within the PVH were positive for phosphorylated TrkB (p-TrkB) (**Figure 2—figure supplement 1F** [\[1\]](#)). Moreover, 20% of p-TrkB⁺ PVH cells express Bdnf, suggesting that Bdnf could have an autocrine mode of action in this cell type (**Figure 2—figure supplement 1F-J** [\[1\]](#)). We also found that *Rai1* is expressed in PVH^{Bdnf} cells that are distinct from larger magnocellular cells such as the PVH oxytocin neurons (**Figure 2—figure supplement 2A-I** [\[1\]](#)). Specifically, 55% of PVH^{Bdnf} neurons express *Rai1*, whereas only 2% of PVH^{Oxytocin} neurons express *Rai1* (**Figure 2—figure supplement 2I** [\[1\]](#)). These data suggest that *Rai1* is enriched in Bdnf-producing neurons in the mouse PVH.

To determine if Bdnf-producing cells depend on *Rai1* to regulate body weight, we deleted one or both copies of *Rai1* using the *Bdnf*^{Cre/+} and the conditional *Rai1*^{fl/fl} mice (Huang *et al.*, 2016 [\[3\]](#)) (**Fig 2A** [\[1\]](#), **Figure 2—figure supplement 3A** [\[1\]](#)). Conditional knockout mice (cKO: *Bdnf*^{Cre/+}; *Rai1*^{fl/fl}) showed a significant reduction in *Rai1* expression in PVH^{Bdnf} neurons compared to controls (**Fig 2B-C** [\[1\]](#), **Figure 2—figure supplement 3B** [\[1\]](#)). The total number of Bdnf-producing cells in the PVH remained similar in Ctrl and cKO mice (**Figure 2—figure supplement 3C** [\[1\]](#)), indicating that *Rai1* loss did not affect PVH neurogenesis or survival.

We monitored body weight weekly and found that male and female cKO mice showed a significant increase in body weight beginning at 15 weeks of age when compared to controls (Ctrl: *Bdnf*^{Cre/+}) (**Fig 2D** [\[1\]](#), **Figure 2—figure supplement 4A** [\[1\]](#)). However, conditional *Rai1* heterozygous mice (*Bdnf*^{Cre/+}; *Rai1*^{fl/+}) did not show a pathological increase in body weight (**Figure 2—figure supplement 3D** [\[1\]](#), **Figure 2—figure supplement 4B** [\[1\]](#)), suggesting that non-Bdnf-producing cells also contribute to SMS pathogenesis. EcoMRI found that 26-week-old cKO male and female mice showed an increased proportion of fat mass but not lean mass (**Fig 2E** [\[1\]](#), **Figure 2—figure supplement 4C** [\[1\]](#)). Further dissection of fat tissues from different organs showed an increased mass of subcutaneous inguinal (S.C. Ing.) fat in cKO mice of both sexes and increased mass of epididymal adipose tissues (eWAT) in female cKOs (**Fig 2F** [\[1\]](#), **Figure 2—figure supplement 4D** [\[1\]](#)).

In contrast, the total mass of brown adipose tissue (BAT) remained unchanged between groups (**Fig 2F** [\[1\]](#), **Figure 2—figure supplement 4D** [\[1\]](#)). In both male and female cKO mice, we found hypertrophic eWAT adipocytes (**Fig 2G** [\[1\]](#), **Figure 2—figure supplement 4E** [\[1\]](#)) and elevated blood leptin levels (**Fig 2H** [\[1\]](#), **Figure 2—figure supplement 4F** [\[1\]](#)). However, there was no significant alteration in blood lipids such as triglycerides (TG), high-density lipoprotein (HDL), low-density lipoproteins (LDL), and very low-density lipoprotein (VLDL) in either male or female cKO mice (**Figure 2—figure supplement 3E-G** [\[1\]](#), **Figure 2—figure supplement 4G-I** [\[1\]](#)). Altogether, these data suggest that obesity caused by ablating *Rai1* from Bdnf-producing cells is associated with increased fat mass, fat deposition, and leptin levels.

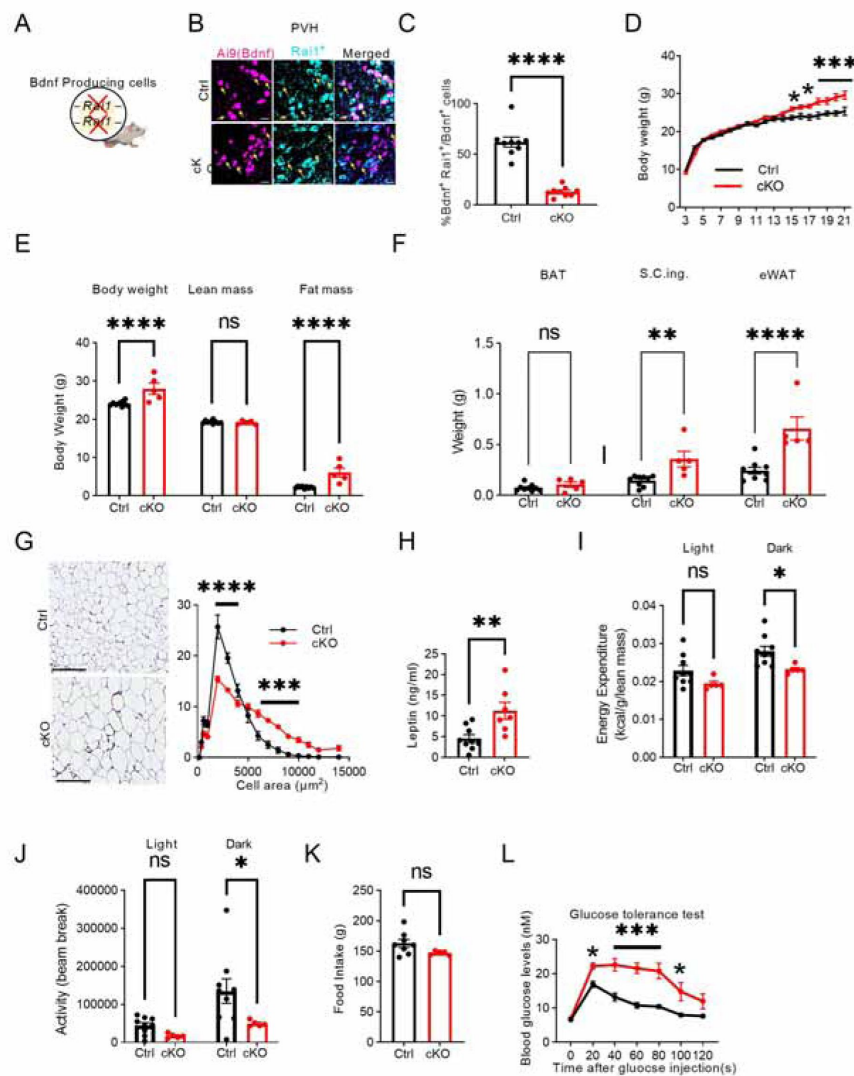


Figure 2

Rai1 is required in Bdnf-producing cells to regulate energy metabolism

- (A) A schematic diagram showing selective deletion of *Rai1* from Bdnf-producing cells in mice.
- (B) Representative images showing that in Ctrl mice (*Bdnf^{Cre/+}; Ai9*), many PVH^{Bdnf} neurons (magenta) express Rai1 (cyan, double-positive cells were indicated with yellow arrowheads). By contrast, PVH^{Bdnf} neurons lack Rai1 expression in the cKO group (yellow arrowheads) (*Bdnf^{Cre/+}; Rai1^{fl/fl}; Ai9*). Scale bars = 20μm.
- (C) Percentage of PVH^{Bdnf} neurons co-expressing Rai1 in Ctrl (n=3) and cKO (n=3) mice. ****p<0.0001, student's t-test.
- (D) Female cKO mice (n = 12) showed a significant weight gain when compared to female Ctrl mice (n = 10).
- (E) Body composition was measured with Echo-MRI, showing an increased fat mass in 26-week-old mice.
- (F) Fat mass of brown adipocytes (BAT), subcutaneous inguinal (S.C.ing) and epididymal white adipose tissue (eWAT) in Ctrl and cKO mice.
- (G) Representative images showing eWAT adipocyte hypertrophy of the cKO mice (left). Scale bar = 500μm. Frequency distribution of adipocytes at each cellular size (right).
- (H) Female cKO mice showed significantly increased blood leptin levels.
- (I) Female cKO mice showed reduced energy expenditure during the dark phase. ns indicates not significantly different.
- (J) Female cKO mice showed reduced locomotor activity during the dark phase. ns indicates difference is not significant.
- (K) Female cKO mice showed similar food intake as control mice. ns indicates not significant.
- (L) Glucose tolerance test showing that cKO mice became hyperglycemic 20 minutes after intraperitoneal glucose administration, suggesting glucose intolerance.

Data are shown as mean±SEM. Statistics for **D-G, I-J, L**: ns indicates not significantly different, *p<0.05, **p < 0.01, ***p<0.001, ****p < 0.0001, one or two-way ANOVA with Šidák's multiple comparisons test.

Obesity could result from increased food intake, reduced energy expenditure, or both. Similar to SMS mice, male and female cKO mice did not have defects in respiratory exchange rates when measured by indirect calorimetry (**Figure 2—figure supplement 3H** [↗](#), **Figure 2—figure supplement 4J** [↗](#)). Interestingly, cKO mice showed sex-specific differences in energy expenditure. During the dark phase, female cKO mice showed a reduction in energy expenditure (**Fig 2I** [↗](#)) and decrease in total locomotor activity (**Fig 2J** [↗](#)). In contrast, total food intake remained unchanged (**Fig 2K** [↗](#)). Male cKO mice became hyperphagic (**Figure 2—figure supplement 4K** [↗](#)), showed increased energy expenditure during the dark phase (**Figure 2—figure supplement 4L** [↗](#)), and exhibited normal locomotor activity when compared to controls (**Figure 2—figure supplement 4M** [↗](#)). These results suggest that obesity in male cKO mice is attributable to overeating despite an increase in energy expenditure.

To determine if *Rai1* loss from *Bdnf*-producing cells impacts metabolic function, a glucose tolerance test (GTT) was performed. Interestingly, cKO mice of both sexes showed normal blood glucose levels right before the test but became hyperglycemic 20 minutes (female cKO) and 45 minutes (male cKO) after glucose administration (**Fig 2L** [↗](#), **Figure 2—figure supplement 4N** [↗](#)). This change was accompanied by a significantly higher area under the curve (AUC) for both groups (**Figure 2—figure supplement 3I** [↗](#), **Figure 2—figure supplement 4O** [↗](#)). While the male cKO mice remained hyperglycemic throughout the GTT (**Figure 2—figure supplement 4P** [↗](#)), the glucose levels in female cKO mice returned to normal towards the end (**Fig 2L** [↗](#)). We also found increased blood insulin levels during the GTT in both sexes (**Figure 2—figure supplement 3J** [↗](#), **Figure 2—figure supplement 4P** [↗](#)). In the insulin tolerance test (ITT), female cKO mice showed moderately elevated glucose levels 20 minutes into the test (**Figure 2—figure supplement 3K** [↗](#)). By contrast, male cKO mice showed considerably higher glucose levels than controls in the ITT (~6.5 ng/ml for cKO and ~2.0 ng/ml for Ctrl) during the 60- to 120-minute intervals (**Figure 2—figure supplement 4Q** [↗](#)). Altogether, these experiments demonstrate that *Bdnf*-producing cells depend on *Rai1* to regulate body weight, adiposity, glucose and insulin tolerance in mice.

***Rai1* deletion reduces PVH^{Bdnf} neuronal excitability**

Bdnf-producing neurons are widely distributed in the hypothalamus, being found in the dorsomedial hypothalamus (Unger *et al.*, 2007 [↗](#)), ventromedial hypothalamus (Xu *et al.*, 2003 [↗](#)), and PVH (An *et al.*, 2015 [↗](#)). To understand how *Rai1* loss affects intrinsic neuronal excitability and synaptic transmission of PVH^{Bdnf} neurons that underlie energy homeostasis (An *et al.*, 2015 [↗](#)), we fluorescently labelled them with Cre-dependent tdTomato (**Fig. 3A** [↗](#)). We found that a hyperpolarized step followed by increasing the holding voltage at resting membrane potentials (Rm) was able to initiate rebound firing or rebound repetitive firing in most PVH^{Bdnf} neurons (**Figure 3—figure supplement 1A** [↗](#)). Hyperpolarization-induced I_H currents and low depolarization events were observed in some PVH^{Bdnf} neurons (**Figure 3—figure supplement 1A** [↗](#)). cKO PVH^{Bdnf} neurons showed reduced action potential (AP) firing frequency at a holding voltage of -60mV compared to Ctrl PVH^{Bdnf} neurons (**Fig 3B-C** [↗](#)). The resting membrane potential (Rm) of cKO (-52mV) neurons showed a 10mV positive shift compared to Ctrl neurons (-62mV) (**Fig 3D** [↗](#)). We also found that fewer cKO PVH^{Bdnf} neurons showed spontaneous firing at Rm (**Fig 3E** [↗](#), Ctrl:10/13; cKO:5/12) and at holding voltage of -70mV (**Fig 3F** [↗](#), Ctrl:6/12; cKO:1/12) compared to Ctrl PVH^{Bdnf} neurons. The threshold of the first AP initiated by a ramp current injection (1 nA/sec) from Rm was significantly increased in cKO neurons (**Fig 3G-H** [↗](#)). Because the higher the holding voltage is, the narrower the shoulder and the lower the amplitude of APs are (**Figure 3—figure supplement 1B** [↗](#)), we compared the AP waveforms at a holding voltage of -60mV, close to the Rm of both genotypes. The AP of the cKO group exhibited a lower peak amplitude (measuring from the threshold to peak) (**Fig 3I** [↗](#)), a normal threshold (**Fig 3J** [↗](#)), and half-width (**Figure 3—figure supplement 1C** [↗](#)). We found that both passive and AP properties of cKO PVH^{Bdnf} neurons were altered. The Rm of cKO neurons shifted to a more positive voltage than the gating threshold of voltage-gated sodium channels (6/12 were more positive than -55mV),

which was unfavourable for their transition from the inactivation state to the closed state. Combined with the higher AP threshold of cKO neurons, these changes contribute to a decreased excitability in cKO PVH^{Bdnf} neurons.

PVN neurons receive glutamatergic and GABAergic inputs and are tonically inhibited by GABAergic neurons (Colmers & Bains, 2018 [↗](#); Gao *et al.*, 2017 [↗](#)). The rightward shift of the probability of miniature inhibitory postsynaptic current (mIPSC) frequency indicated that cKO PVH^{Bdnf} neurons received a higher frequency of inhibitory inputs, although the average frequency (Fig 3K [↗](#)) was not significantly different between groups. The probability of mIPSC amplitude also showed a right shift, suggesting an increased mIPSC postsynaptic response (Fig 3L [↗](#), Figure 3—figure supplement 1D [↗](#)). Furthermore, we observe an increased area under the curve (Fig 3M [↗](#)), indicating that *Rai1* loss induced a larger total outward current upon the inhibitory inputs. Interestingly, we found that cKO PVH^{Bdnf} neurons had a smaller cellular diameter (Figure 3—figure supplement 1E [↗](#)), indicating that the increased inhibitory currents were unlikely due to changes in passive membrane properties associated with increased cell size. Overall, our data argue that *Rai1* deletion decreases the excitability of PVH^{Bdnf} neurons and likely increases pre-synaptic inhibitory transmission and post-synaptic inhibitory responses, resulting in decreased AP firing.

Rai1 expression is necessary for PVH^{Bdnf} neurons to regulate energy homeostasis

Reduced PVH^{Bdnf} neuronal activity in cKO mice suggested that *Rai1* loss in Bdnf-producing PVH neurons might be sufficient to induce defective energy homeostasis *in vivo*. To test this hypothesis by deleting *Rai1* only in PVH^{Bdnf} neurons but not other Bdnf-producing cells, we performed PVH^{Bdnf}-specific *Rai1* deletion using a CRISPR/Cas9 system packaged into AAV9. Two viruses were delivered; one encoding a reporter molecule (GFP) and three single guide RNAs targeting the *Rai1* gene (sgRai1), and the other carrying Cre-dependent *Staphylococcus aureus* CRISPR-associated protein 9 (saCas9) (Fig 4A [↗](#)). Hereafter, our experiments focus on female SMS mice since they showed more consistent defects in fat distribution (Burns *et al.*, 2010 [↗](#)) and body weight regulation (Huang *et al.*, 2016 [↗](#)) than males. Bilateral administration of two AAVs into the PVH of female *Bdnf*^{Cre/+} mice (Exp group) resulted in a significant loss of *Rai1* protein expression in PVH^{Bdnf} neurons when compared to *Bdnf*^{Cre/+} mice injected with an AAV encoding the sgRai1 and GFP (Ctrl group) (Fig 4B–C [↗](#), Figure 4—figure supplement 1A [↗](#)). Quantification of AAV-infected (GFP⁺) Bdnf neurons showed that while the total number of Bdnf⁺ (Figure 4—figure supplement 1B [↗](#)) and Bdnf⁺ GFP⁺ (Figure 4—figure supplement 1C [↗](#)) cells was not different in the two groups, the percentage of AAV-infected Bdnf⁺ cells that express *Rai1* was significantly reduced in the Exp group (Fig 4C [↗](#), Figure 4—figure supplement 1D–E [↗](#)). Deleting *Rai1* from the PVH^{Bdnf} neurons resulted in a significant increase in body weight beginning at 17 weeks of age (Fig 4D [↗](#)). Similar to the cKO (*Bdnf*^{Cre/+}; *Rai1*^{fl/fl}) mice, obesity in 26-week-old Exp mice was due to increased fat mass (Fig 4E [↗](#)) and more specifically, increased subcutaneous inguinal (S.C. ing) and epididymal white adipose tissues (eWAT) but not BAT (Fig 4F [↗](#)). We also found that eWAT adipocytes became hypertrophic (Fig 4G [↗](#)). Similar to the cKO mice, food intake did not differ between Exp and Ctrl groups (Fig 4H [↗](#)), likely due to the high level of satiety signals in the brain caused by increased blood leptin levels (Fig 4I [↗](#)). We also found that energy expenditure (Fig 4J [↗](#)), respiratory exchange rate (Figure 4—figure supplement 1F [↗](#)), and blood lipid parameters, including TG (Figure 4—figure supplement 1G [↗](#)), HDL (Figure 4—figure supplement 1H [↗](#)), LDL and VLDL levels (Figure 4—figure supplement 1I [↗](#)), were similar in Exp and Ctrl groups, consistent with the cKO data (Figure 4—figure supplement 14 [↗](#)). Interestingly, Exp mice showed significantly less locomotor activity during the dark phase than the Ctrl mice (Fig 4K [↗](#)). A GTT indicated that Exp mice have a hyperglycemic profile (Fig 4L [↗](#)), accompanied by a slightly higher area under the curve for blood glucose levels (*p*=0.05) (Fig 4M [↗](#)). Insulin levels in Exp mice were also significantly elevated 15 minutes after glucose administration (Fig 4N [↗](#)).

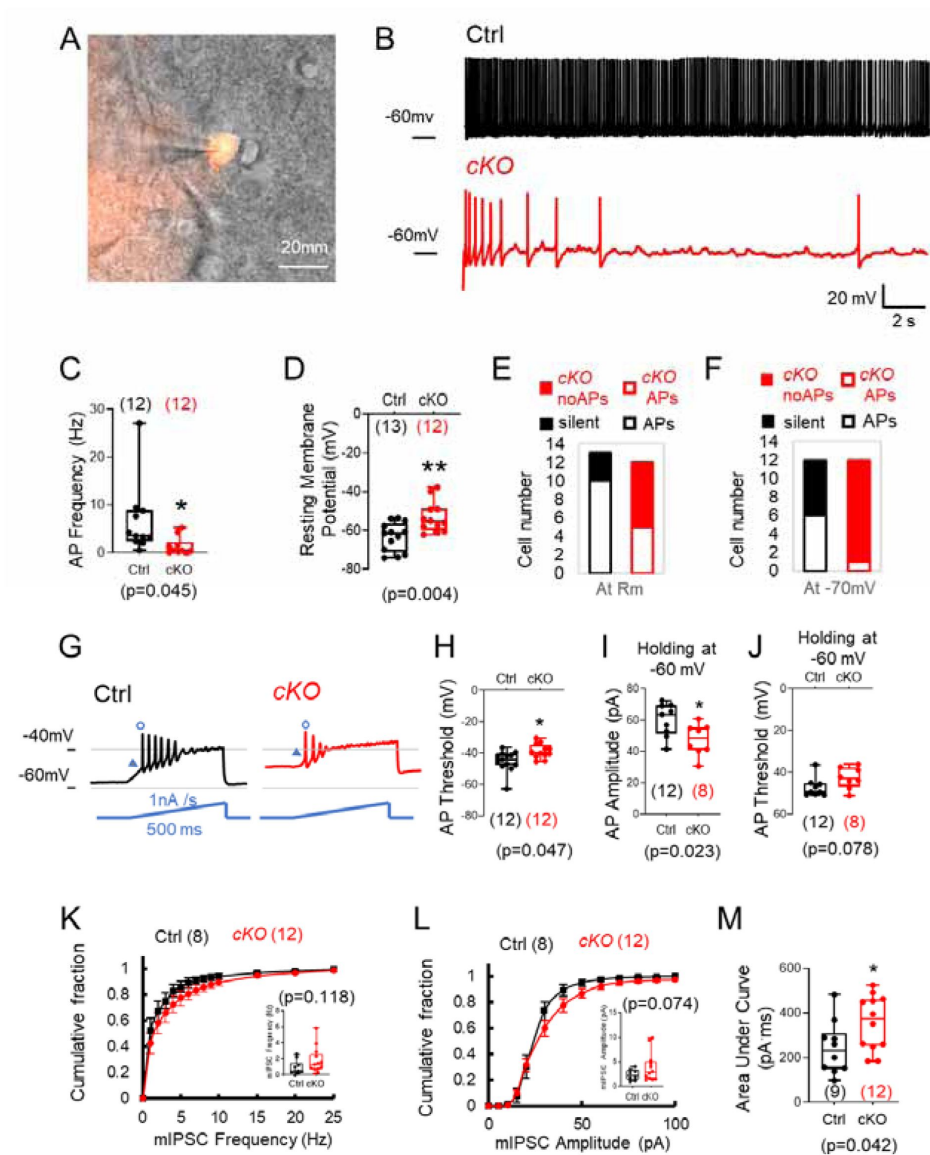


Figure 3

Rai1 loss reduces intrinsic neuronal excitability and enhances inhibitory synaptic transmission of PVH^{Bdnf} neurons

(A) A representative image showing a patched PVH^{Bdnf} neuron labelled by *Bdnf*^{Cre}-dependent tdTomato fluorescence signals.

(B-C) Representative traces of spontaneous firing of control (black) and cKO (red) neurons at a holding voltage of -60mV **(B)**. The average spontaneous AP firing frequencies are shown in **(C)**.

(D) The resting membrane potentials of control (black) and cKO (red) PVH^{Bdnf} neurons.

(E-F) The number of control (black) and cKO (red) PVH^{Bdnf} neurons that showed spontaneous firing (solid) or are silent (open) at resting membrane potentials **(e)** and at a holding voltage of -70mV **(f)**.

(G) Representative traces of elicited APs in control (black) and cKO (red) PVH^{Bdnf} neurons responding to an inject current (bottom, blue), which ramps up from the resting membrane potentials at 1 nA per second for 500ms.

(H) The threshold of the first AP initiated by a current ramp in control (black) and cKO (red) PVH^{Bdnf} neurons.

(I-J) The amplitude **(I)** and threshold **(J)** for the first AP firing at the holding voltage of -60mV in control (black) and cKO (red) PVH^{Bdnf} neurons.

(K) The cumulative fractions of mIPSC frequency measured in control (black) and cKO (red) PVH^{Bdnf} neurons. The average frequency of events from a 2-min stable recording of each neuron is shown in the inset.

(L) The cumulative fractions of mIPSC amplitude of control (black) and cKO (red) PVH^{Bdnf} neurons. The average amplitude of events from a 2-min stable recording of each neuron is normalized to its capacitance and shown in the inset.

(M) The area under the curve of each mIPSC event from a 2-min recording is averaged. Controls are shown in black, and cKOs are shown in red. Data are shown as mean±SEM. Statistics: Neuron numbers are in brackets (more than 3 mice/genotype); *p<0.05, unpaired student's t-tests.

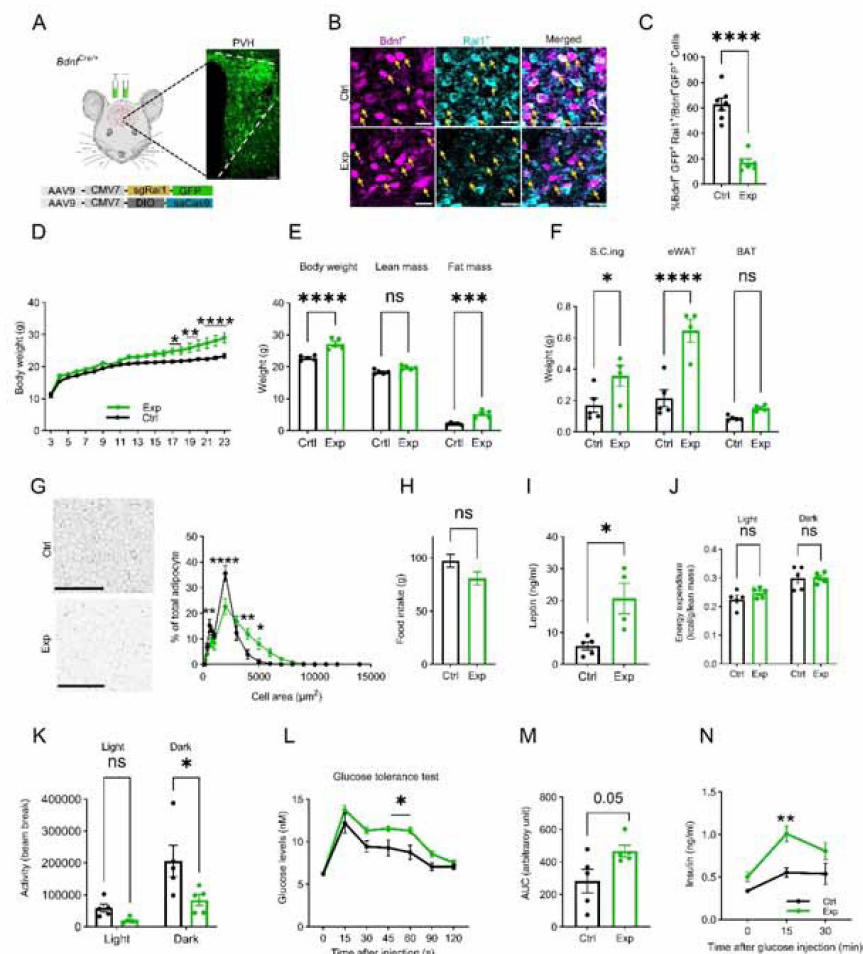


Figure 4

Selective *Rai1* ablation in PVH^{Bdnf} neurons induces obesity

(A) A schematic showing stereotaxic injection of adeno-associated viruses (AAVs) expressing Cas9 protein and single-guide RNAs targeting *Rai1* (sgRai1) to delete *Rai1* from the PVH^{Bdnf} neurons of *Bdnf*^{Cre/+} female mice. Scale bar = 50 μ m.

(B) Representative images showing that *Rai1* immunoreactivity (cyan) was dramatically reduced in *Bdnf*^{Cre/+}; *Ai9* mice injected with both sgRai1 and DIO-saCas9 viruses (Exp group, bottom row) into the PVH. In contrast, many PVH^{Bdnf} neurons in *Bdnf*^{Cre/+}; *Ai9* mice injected only with the sgRai1 virus showed *Rai1* expression (Ctrl group, top row).

(C) The percentage of PVH^{Bdnf} neurons co-expressing *Rai1* and GFP (virus) was significantly reduced in the Exp group, indicating a successful *Rai1* deletion.

(D) PVH^{Bdnf} neuron-specific *Rai1* deletion induced body weight gain in Exp mice.

(E) Body composition was measured with Echo-MRI, showing an increased fat mass deposition in 26-week-old Exp mice.

(F) Fat deposition analysis shows that the Exp group has significantly more subcutaneous inguinal (S.C.ing) and epididymal white adipose tissue (eWAT) mass (grams). ns indicates not significantly different.

(G) Representative images showing eWAT adipocyte hypertrophy of the Exp group (left). Scale bar = 500 μ m. Frequency distribution of adipocytes at each cellular size (right). ns indicates not significantly different.

(H) Exp and control mice showed similar food intake. ns indicates not significant, student's t-test.

(I) Blood leptin levels were significantly increased in the Exp mice. *p < 0.05, student's t-test.

(J) Exp and control mice showed similar energy expenditure.

(K) Exp mice showed reduced locomotor activity in the dark phase.

(L) Exp mice became hyperglycemic during the glucose tolerance test.

(M) Measurement of the area under the curve (AUC) in Exp and Ctrl mice during the glucose tolerance test. p = 0.0522, student's t-test.

(N) Exp mice showed increased plasma insulin levels during the glucose tolerance test.

Data are shown as mean $\pm$ SEM. Statistics for **C-G, J-L, N**: ns indicates not significantly different, *p < 0.05, **p < 0.01, ***p < 0.001, ****p < 0.0001, one or two-way ANOVA with Šidák's multiple comparisons test.

While Exp mice showed a trend towards increased glucose levels in insulin tolerance tests, the change was not statistically different (**Figure 4—figure supplement 1J** [↗](#)). Altogether, these experiments indicate that conditional deletion of *Rai1* from PVH^{Bdnf} neurons during adolescence impairs the regulation of body weight, adiposity, glucose tolerance, and locomotion. These changes recapitulate most phenotypes observed in cKO mice with *Rai1* deleted in Bdnf-producing cells from early development.

Pharmacologically enhancing TrkB signalling partially alleviates obesity in SMS mice

Our findings so far indicate that multiple pathways, including neurotrophin signalling, are dysregulated in SMS mice, and that *Rai1* loss from Bdnf-producing cells, including PVH^{Bdnf} neurons, impairs energy homeostasis *in vivo*. Next we sought to examine the contribution of Bdnf-TrkB dysfunction in SMS pathogenesis by pharmacologically activating downstream targets of the Bdnf-TrkB pathway.

We systematically treated mice with LM22A-4, a small molecule Bdnf loop domain mimetic (Massa *et al.*, 2010 [↗](#)). To measure the bioavailability of LM22A-4 in the SMS brain, we simultaneously administered LM22A-4 intranasally (5 mg/kg body weight) and intraperitoneally (50 mg/kg body weight), using the highest doses used in previous studies (Canals *et al.*, 2004 [↗](#); Chang *et al.*, 2006 [↗](#); Gu *et al.*, 2022 [↗](#); Kron *et al.*, 2012 [↗](#); Ogier *et al.*, 2007 [↗](#)). We then harvested forebrain and hypothalamic tissues one- or three-hours after drug administration. Spectrometric analysis found a significantly higher LM22A-4 concentration in both brain regions one-hour post-treatment, while its concentration decreased three hours after treatment (**Figure 5—figure supplement 1A** [↗](#)). Consistent with the reduced neurotrophin signalling observed in our proteomic analysis, saline-treated SMS mice showed reduced phosphorylated-Akt (p-Akt; **Fig 4A** [↗](#), **Figure 5—figure supplement 1B** [↗](#)) levels in the hypothalamus. LM22A-4 treatment significantly increased the levels of p-Akt in the SMS mice 1-hr post-treatment (**Fig 4A** [↗](#), **Figure 5—figure supplement 1B** [↗](#)), indicating increased signalling downstream of the Bdnf-TrkB pathway.

We next examined if chronic LM22A-4 treatment was sufficient to reverse or modify the progression of the obesity phenotype in adult SMS mice. Specifically, we treated SMS mice daily from 8 to 14 weeks of age, during which SMS mice began to gain significantly more weight. Saline-treated SMS mice (**Fig 5C** [↗](#)) were significantly heavier than saline-treated control mice from 9 weeks of age onward. Interestingly, LM22A-4-treated SMS mice showed a similar body weight to saline-treated control littermates until 13 weeks of age but became obese by 14 weeks of age (**Fig 5D** [↗](#)). Consistent with a partial rescue, LM22A-4-treated SMS mice were significantly less obese than saline-treated SMS mice throughout the treatment period (**Fig 5E** [↗](#)). Control mice treated with LM22A-4 did not show a further reduction in body weight (**Figure 5—figure supplement 1C** [↗](#)), indicating that the therapeutic effects were specific to SMS pathogenesis.

We measured food intake and found that LM22A-4 did not significantly rescue increased food intake in SMS mice. (**Fig 5F** [↗](#)). SMS mice showed normal energy expenditure (**Figure 5—figure supplement 1D** [↗](#)) and respiratory exchange rate (**Figure 5—figure supplement 1E** [↗](#)), neither of which was altered by LM22A-4 treatment. Interestingly, we did observe a partial rescue of locomotor activity in LM22A-4-treated SMS mice that was not significantly different from saline-treated Ctrl or SMS mice (**Fig 5G** [↗](#)). We also found that LM22A-4 treatment partially rescued fat deposition (**Fig 5H** [↗](#)) and completely rescued HDL levels (**Fig 5I** [↗](#)) in SMS mice. During an insulin tolerance test, we found that saline-treated SMS mice showed increased baseline insulin levels (**Fig 5J** [↗](#)) and blood glucose levels during the test (**Fig 5K** [↗](#)), both of which were partially rescued by LM22A-4 treatment.

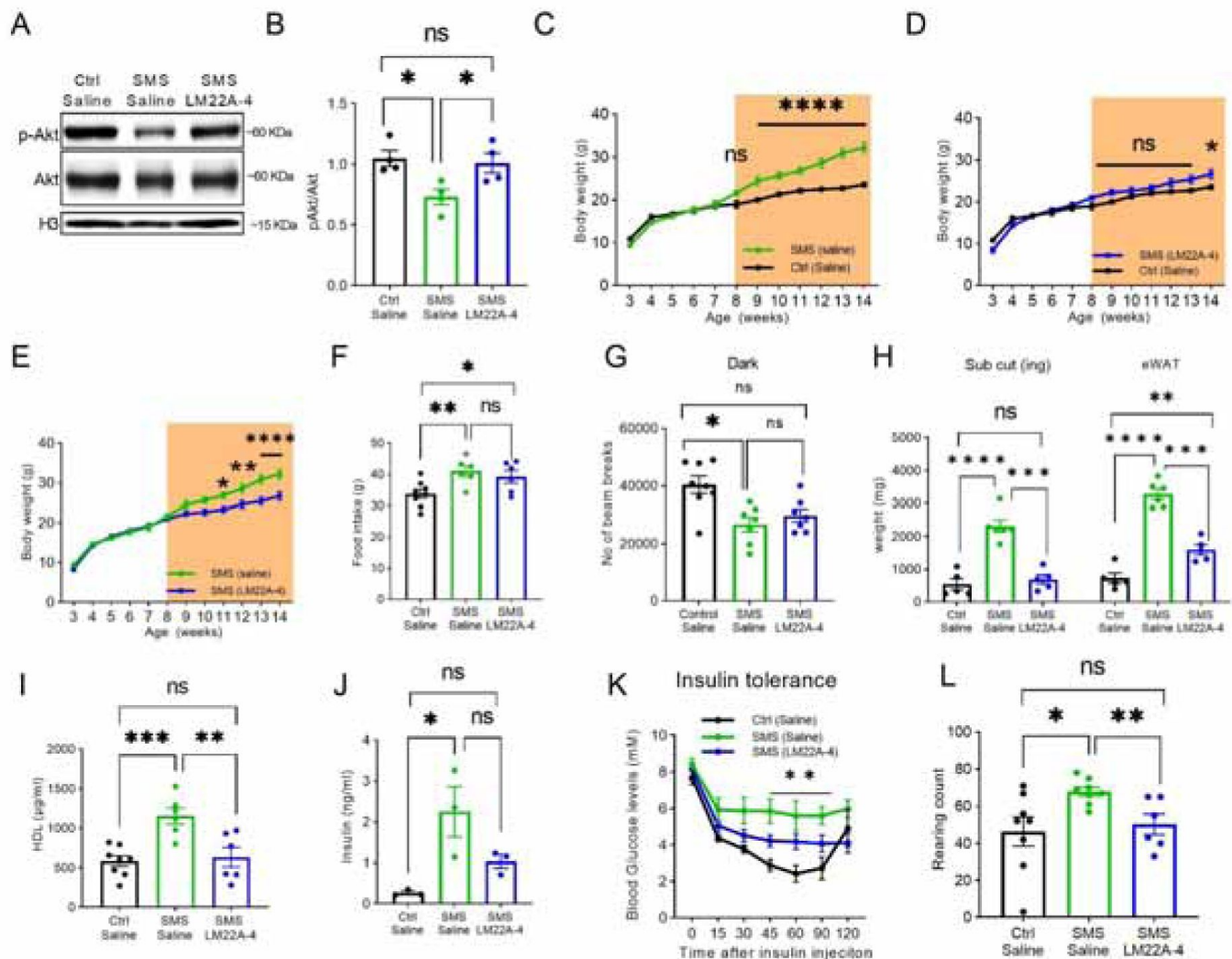


Figure 5

LM22A-4 treatment partially alleviates obesity and stereotypic behaviour in adult SMS mice

- (A)** Representative western blot showing p-Akt, total Akt, Histone 3 (H3, a loading control) levels in the hypothalamus of saline-treated Ctrl, saline-treated SMS and LM22A-4-treated SMS mice.
- (B)** Quantification showing phosphorylation deficits in Akt protein levels of the hypothalamus were reversed by LM22A-4 treatment in SMS mice.
- (C-E)** Body weight of saline-injected control, saline-injected SMS, and LM22A-4 injected SMS mice. Saline or LM22A-4 treatment periods were highlighted in orange. Ns indicates not significantly different.
- (F)** Food intake (in grams) of saline-injected control, saline-injected SMS, and LM22A-4 injected SMS mice.
- (G)** Locomotor activity (number of beam breaks) for saline-injected control, saline-injected SMS, and LM22A-4 injected SMS mice.
- (H)** Fat deposition analysis showing that the saline-injected SMS group has significantly more subcutaneous inguinal (S.C.ing) and epididymal white adipose tissue (eWAT) mass (grams) than saline-injected Ctrl mice. S.C.ing and eWAT mass in LM22A-4 injected mice is comparable to the saline-injected mice.
- (I)** Insulin tolerance test performed on saline-injected control, saline-injected SMS, and LM22A-4 injected SMS mice, two-week post daily administration of LM22A-4 or saline, shows reductions in the blood glucose levels of LM22A-4 treated SMS mice. * Shows significance between saline injected Ctrl and saline injected SMS group.
- (J)** Insulin levels in saline-injected control, saline-injected SMS, and LM22A-4 injected SMS mice right before the start of insulin tolerance test (time point 0).
- (K)** High density lipoprotein (HDL) levels were restored to normal in LM22A-4 injected SMS mice.
- (L)** Stereotypic rearing behaviour was fully rescued in LM22A-4 treated SMS mice.

Data are shown as mean±SEM. Statistics for **B-L**: ns indicates not significantly different, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$, one or two-way ANOVA with Šidák's multiple comparisons test.

In addition to a deficit in energy homeostasis, SMS mice showed two highly reproducible behavioral deficits: increased stereotypic rearing in the open field and decreased social dominance in the tube test (Huang *et al.*, 2018 [DOI](#); Rao *et al.*, 2017 [DOI](#)). Saline-treated SMS mice showed significantly more rearing behavior when compared to the saline-treated controls (Fig 5L [DOI](#)). Interestingly, LM22A-4 treatment reduced the repetitive rearing behaviour in SMS mice (Fig 5L [DOI](#)). LM22A-4 treatment did not change repetitive behaviors in control mice (Figure 5—figure supplement 1F [DOI](#)), suggesting the effect is specific to *Rai1* haploinsufficiency. However, we did not observe a rescue of social dominance deficits in SMS mice (Figure 5—figure supplement 2A-E [DOI](#)). Together, these data indicate that LM22A-4 treatment delayed the onset of obesity and fully rescued repetitive behaviour in SMS mice, suggesting that Bdnf-TrkB signalling dysfunction contributes to changes in the mouse SMS brain linked to obesity and behavioral phenotypes.

Discussion

The hypothalamus, and Bdnf-TrkB signalling in this region, control body weight by regulating the intricate balance between energy intake and expenditure (Xu & Xie, 2016 [DOI](#)). The role of Bdnf-producing neurons in human obesity disorders and how the activity of these cells is regulated remained poorly studied. We found that *Rai1* deficiency, linked to Smith-Magenis syndrome (SMS), regulates Bdnf expression and maintains intrinsic excitability and inhibitory synaptic transmissions in PVH^{Bdnf} neurons.

Consistent with this mechanism, we found that removing *Rai1* from Bdnf-producing neurons or PVH^{Bdnf} neurons results in imbalanced energy homeostasis and results in obesity. A small molecule drug LM22A-4 led to increased Akt phosphorylation and partially rescued locomotor activity, insulin intolerance, and HDL levels, while delaying the onset of obesity in SMS mice. These findings highlight a potential role for hypothalamic Bdnf-producing neurons, specifically PVH^{Bdnf} neurons, in SMS pathogenesis and highlight Bdnf signalling as a potentially druggable pathway to improve energy homeostasis defects and repetitive behaviours commonly observed in SMS patients (Burns *et al.*, 2010 [DOI](#); Moss *et al.*, 2009 [DOI](#)).

The involvement of different PVH subtypes in human obesity is only beginning to be elucidated. Dysfunction of oxytocin PVH neurons is linked to the Prader-Willi syndrome (Lee *et al.*, 2012 [DOI](#)), another genetic disorder associated with obesity. Since *Rai1* expression is enriched in PVH^{Bdnf} neurons but not PVH^{Oxytocin} neurons, our data suggest that distinct pathophysiological mechanisms underlie SMS and Prader-Willi syndrome.

Conditional knockout analysis allowed us to specifically map the cell-specific origin of neuropathological phenotypes, such as SMS-like symptoms in mice. Our previous work found that deleting *Rai1* from cortical excitatory neurons using an *Emx1*^{Cre} allele (Gorski *et al.*, 2002 [DOI](#)), from the GABAergic neurons using a *Gad2*^{Cre} allele (Taniguchi *et al.*, 2011 [DOI](#)), or from astrocytes using a *GFAP*^{Cre} allele (Garcia *et al.*, 2004 [DOI](#)) did not induce SMS-like obesity (Huang *et al.*, 2016 [DOI](#)). In contrast, obesity could be induced by ablating *Rai1* from Sim1-lineage neurons and, to a lesser extent, SF1-lineage neurons (Huang *et al.*, 2016 [DOI](#)). This analysis suggests that *Rai1* expression in PVH is more important than in ventromedial nucleus of the hypothalamus (VMH) for body weight regulation. Interestingly, embryonic deletion of *Bdnf* from SF1-lineage neurons including the VMH only resulted in modest weight gain (Unger *et al.*, 2007 [DOI](#)). By contrast, deleting *Bdnf* from Sim1-lineage neurons including the PVH resulted in a more severe obesity phenotype (An *et al.*, 2015 [DOI](#)). This is consistent with the idea that Bdnf expression in PVH is critical for energy homeostasis. We found that *Rai1* expression in Bdnf-producing cells regulates energy homeostasis in a sexually dimorphic manner. Specifically, female cKO mice showed normal food intake, decreased energy expenditure, and reduced locomotor activity. In contrast, male cKO mice showed increased food intake with no changes in locomotor activity. Intriguingly, energy expenditure was increased in male cKO mice, likely due to the higher demand to maintain normal activity levels in obese mice.

In both sexes, cKO mice showed an increased fat mass and adiposity with no change in lean mass, highlighting an important role for *Rai1* in regulating fat deposition. Our data also indicate that Bdnf-producing cells are only partially responsible for obesity in SMS: male and female cKO mice become overweight by 15 weeks, in contrast to SMS mice which become obese by 9 weeks of age, suggesting that non-Bdnf neurons contribute to SMS phenotypes.

We found the metabolic phenotypes induced by deleting *Rai1* from Bdnf-producing cells were specifically due to *Rai1* deletion in PVH^{Bdnf} neurons. Therefore, our study focused on PVH^{Bdnf} neurons. Electrophysiological experiments showed that most wild-type PVH^{Bdnf} neurons display high frequency spontaneous AP firing (1-10 Hz) at the resting membrane potential. The majority of PVH^{Bdnf} neurons fire sharp APs held at a hyperpolarized potential and show repetitive overshoot during the current ramp test. Loss of *Rai1* significantly reduced intrinsic neuronal excitability and increased inhibitory synaptic transmissions onto PVH^{Bdnf} neurons. Because GABAergic *Rai1* loss did not induce obesity (Huang *et al.*, 2016 [\[16\]](#)) and most Bdnf-producing neurons are glutamatergic (Canals *et al.*, 2001 [\[17\]](#)), the increased mIPSC amplitude was likely due to postsynaptic changes in GABA receptor functions caused by *Rai1* loss. The ionic changes in PVH^{Bdnf} neurons that precisely explain these excitability changes await future investigations. Different groups of PVH neurons show phasic bursting or slowly inactivating delayed rectifier potassium conductance, which are properties mediated by different potassium channels (Lee *et al.*, 2012 [\[18\]](#)). Given that our previous RNA-seq experiment found that hypothalamic *Rai1* loss induces misexpression of potassium channel subunits *Kcnc2* and *Kcnj11* (Huang *et al.*, 2016 [\[16\]](#)), *Rai1*-dependent potassium channels are likely involved in altered neuronal activity. Furthermore, we found fewer low threshold depolarization events in cKO PVH^{Bdnf} neurons during the hyperpolarized voltage step, suggesting a potential involvement of HCN channels.

Our previous work found that *Rai1* loss in the dentate gyrus resulted in neuronal hyperexcitability (Chang *et al.*, 2022b [\[19\]](#)). Here we demonstrated that *Rai1* loss in the hypothalamus resulted in neuronal hypoactivity, suggesting that *Rai1* regulates neuronal excitability in a cell type-specific manner. Reduced PVH^{Bdnf} neuronal activity upon *Rai1* loss could potentially explain obesity, consistent with a previous report that found chemogenetic activation of PVH^{Bdnf} neurons promotes negative energy balance by decreasing feeding and increasing energy expenditure (Wu & Xu, 2022 [\[20\]](#)).

In future work, it would be interesting to dissect the sex-specific role of *Rai1* in regulating the functional development of PVH^{Bdnf} neurons. In addition to a role for *Rai1* in regulating neuronal activity, it is possible that *Rai1* regulates other aspects of PVH^{Bdnf} neuronal function, including projections to other brain regions to regulate energy homeostasis. In this regard, it would be interesting to dissect how *Rai1* controls circuit connectivity of PVH^{Bdnf} neurons and how *Rai1* loss in PVH^{Bdnf} neurons affects nearby PVH^{TrkB} neurons that receive Bdnf signaling and are linked to body weight regulation (An *et al.*, 2020 [\[21\]](#)).

At the molecular level, our RPPA data show that *Rai1* depletion disrupted multiple intracellular signalling pathways and resulted in the hypoactivation of specific phospho-signalling proteins. Multiple early studies support the notion that *Rai1* directly promotes Bdnf expression. First, hypothalamic *Bdnf* mRNA levels were reduced in SMS mice (Burns *et al.*, 2010 [\[22\]](#); Javed *et al.*, 2021 [\[23\]](#)) or mice with *Rai1* conditional deletions (Huang *et al.*, 2016 [\[16\]](#)). Second, *Rai1* protein directly binds to a promoter region of *Bdnf* (Huang *et al.*, 2016 [\[16\]](#)). Finally, when *Rai1* expression was induced using CRISPR activation, *Bdnf* mRNA levels were significantly elevated (Chang *et al.*, 2022a [\[24\]](#)).

To lend further support, here we show that SMS mice have reduced hypothalamic Bdnf protein levels, as well as reduced activation of a TrkB downstream target Akt. Reduced Akt activity is frequently associated with insulin resistance and obesity in mice (Shao *et al.*, 2000 [\[25\]](#)) and children (Su *et al.*, 2021 [\[26\]](#)). We also found that obesity, increased HDL levels, and insulin insensitivity in

SMS mice could be partially ameliorated using a small molecule drug (LM22A-4) that promotes Akt phosphorylation. LM22A-4 penetrates the blood-brain barrier reasonably well and has been shown to ameliorate symptoms of other neurological disorders associated with Bdnf dysfunction including Rett syndrome, Huntington's disease, Dravet syndrome, and refractory neonatal seizures (Canals *et al.*, 2004 [↗](#); Chang *et al.*, 2006 [↗](#); Gu *et al.*, 2022 [↗](#); Kron *et al.*, 2012 [↗](#); Ogier *et al.*, 2007 [↗](#)). The partial rescue of SMS phenotypes that we observed contrasts with our previous findings that found (1) AAV-mediated Bdnf overexpression in PVH at 3 weeks of age and (2) genetically overexpressing Bdnf from early development could fully prevent the development of obesity in SMS mice (Javed *et al.*, 2021 [↗](#)). It is therefore possible that further increasing TrkB signalling activation and/or earlier intervention might improve therapeutic outcomes.

Alternatively, our RPPA data showed that in addition to disruption of neurotrophin signalling, multiple pathways including mTOR were downregulated in SMS. Increasing hypothalamic mTOR activity has been shown to decrease food intake and body weight (Cota *et al.*, 2006 [↗](#)). Therefore, it is possible that enhancing both mTOR and TrkB signalling could achieve stronger rescue of obesity in SMS mice. Nevertheless, these data support the concept that drugging TrkB signalling may improve stereotypical repetitive behavior and partially alleviate metabolic functions in a mouse model of SMS. Delayed symptom onset could provide a window of opportunity for other interventions to further modify disease progression, an area requiring further investigation.

This study has several limitations. First, whether the PVH^{Bdnf} neurons depend on Rai1 to regulate adaptive thermogenesis awaits future analysis. It is known that a subset of posterior PVH^{Bdnf} neurons project to the spinal cord and regulate thermogenesis (An *et al.*, 2015 [↗](#)). However, we found the weights of brown adipocytes were similar in control and cKO mice. Moreover, SMS mice have normal respiratory exchange rate and there are currently no reports of defective adaptive thermogenesis in SMS patients. This suggests that adaptive thermogenesis is not impacted in SMS. Another limitation is that we did not test different doses, durations, and time points for LM22A-4 treatment. The precise mechanisms by which LM22A-4 improves body weight, metabolic function, and repetitive behaviours also remain unknown. Nevertheless, our work identifies Rai1 as a novel regulator of the neuronal excitability of PVH^{Bdnf} neurons. This regulation is critical for regulating food intake and energy expenditure. Our evidence suggests that Bdnf-producing neurons are involved in SMS pathogenesis and that enhancing Bdnf-TrkB signalling should be explored as a future therapeutic avenue for obesity and metabolic conditions associated with reduced neurotrophin signalling (Xu & Xie, 2016 [↗](#)).

Materials and Methods

Animal Management

The animal protocol was approved by the animal care committee of the Montreal General Hospital and was performed according to the guidelines of the Canadian Council on Animal Care. Mice were group-housed in 12-hour light/12-hour dark cycles with an *ad libitum* supply of regular chow and water. The exception was in the fasting studies when the animals were food deprived for 5 hours and in CALMS and food intake studies when animals were single-housed in a 12-hour light/12-hour dark cycle with an *ad libitum* supply of regular chow and water. All mice were maintained in a C57Bl/6J background. *Bdnf*^{Cre/Cre}, *Rosa-tomato*^{td/t}(Ai9), and *Actβ*^{Cre/Cre} mice were purchased from the Jackson laboratory. The *Rai1*^{fl/fl} strain was previously generated and characterized by us (Huang *et al.*, 2016 [↗](#)). SMS mice (*Rai1*^{+/-}: whole-body germline *Rai1* heterozygous knockout that genetically mimics SMS) were generated by crossing germline *Actβ*^{Cre/Cre} mice with *Rai1*^{fl/+}. To generate conditional knockout mice, *Bdnf*^{Cre/+}; *Rai1*^{fl/+} mice were crossed with *Rai1*^{fl/+} to create control and experimental groups from the same littermate.

Metabolic profiling

Metabolic profiling was conducted in three different experimental conditions: male and female conditional knockout mice, female virus-injected *Bdnf*^{Cre/+} mice, and female drug- or saline-injected SMS and control mice. Metabolic profiling included body weight and body composition analysis, food intake measurement, indirect calorimetry for locomotion activity, energy expenditure (EE), respiratory exchange rate (RER), measurement of serum lipids and leptin levels, adipose tissue histology analysis, and glucose and insulin tolerance tests. All experiments were performed as previously conducted experiments (Javed *et al.*, 2021 [DOI](#)). A brief description is detailed below.

Body weight and body composition analysis

The body weight of animals was measured weekly from week 3 (weaning age) onward. Body composition, including total fat and lean mass, were assessed using a nuclear EchoMRI whole-body analyzer. Toward the end of experiments, animals were euthanized to collect brown adipose, visceral, subcutaneous (inguinal) and ependymal white adipose tissues and weighed using an analytical scale (Sartorius).

Food Intake Measurement

Mice were single-housed for food intake measurement. Food intake was measured over a one-week period during which food was weighed prior to placement in the cage with minimal bedding. Each morning remaining food inside the cage was weighed. The amount of total food consumed was averaged over 7 days.

Indirect Calorimetry

A Comprehensive Lab Animal Monitoring System (CLAM) (Columbus instruments) was used to measure the RER and EE (normalized over lean mass). Animals were single-housed in the CLAM apparatus at 21°C (70°F) in a light–dark cycle matching their housing conditions for 24 hours (acclimation), followed by 48 hours of measurement.

Serum lipid level determination

A fluorescence-based assay kit (Abcam ab65390) was used to measure serum lipid levels. The manufacturer's instructions were followed except that we used half-area black 96-well microplates. Fluorescence measurements at 535/587 nm (Ex/Em) were recorded with the Ensign instrument from PerkinElmer. A new standard curve ranging from 0 to 10 µg/ml was produced for each microplate to accurately determine lipid levels in serum samples. Standards and samples were loaded and analyzed twice.

Histology of adipose tissue

Ependymal white adipose tissues (WAT) were excised immediately after the animals were euthanized and fixed with 10% formalin in PBS. The fixed tissues were dehydrated, embedded in paraffin, and sectioned. Two 5-micrometre thick sections were taken from each animal and stained with hematoxylin and eosin. Images of sections were then captured using an Aperio ScanScope CS slide scanner at 20X optical magnification, and the resulting images were saved in jpeg format using the ImageScope software. The images were manually cleaned to remove non-adipose tissues such as lymph nodes. An automated method using the Adiposoft plugin version 1.16 was utilized to determine adipocyte size and number from the jpeg images in Fiji version 2.1.0/1.53c. The Adiposoft settings used excluded cells on edges and set a minimum and maximum cell diameter of 20 and 300 µm, respectively. An experimental value of 2.49 µm per pixel was determined using the Aperio scale bar and used for Adiposoft. To ensure accuracy, two images

from each animal were analyzed blindly, examining more than 1600 cells per mouse and more than 9100 cells per condition. Adipocyte size frequencies were calculated using the frequency function in Excel.

Glucose and insulin tolerance tests

For the glucose tolerance test, experimental mice were food deprived for 5 hours with *ad libitum* water supply. A bolus of glucose (1.5g/kg of lean body weight) was then administered intraperitoneally and glycemia was measured from tail vein blood samples using the Accu-chek Performa glucometer at T0 (before the injection) as well as at 15, 30, 45, 60, 90 and 120 min (after the injection). Tail vein blood samples were collected via a capillary for insulin assays at T0, 15 and 30 min. For insulin tolerance tests, experimental mice were food-deprived for 5 hours with *ad libitum* access to water. A bolus of insulin (1 U/kg of lean body weight) was administered via an IP injection and glycemia was measured from blood sampled at the tail vein using a Accu-chek Performa glucometer at T0 (before the injection), as well as at 15, 30, 45, 60, 90 and 120 min (after the injection). Tail vein blood samples were collected via a capillary for insulin assays at T0.

Stereotaxic Injections

Bdnf^{Cre/+} mice were anesthetized using isoflurane, and the ears were barred onto a stereotaxic device (David Kopf Instruments). Adeno-associated viruses (AAVs, volume = 250 nl) were bilaterally injected into the PVH using the following coordinates relative to the Bregma: -0.6 mm anteroposterior (AP), ± 0.35 mm mediolateral (ML) and -5.52 mm dorsoventral (DV). A description of the AAV constructs and titre used is detailed below:

AAV9-CMV7-DIO-saCas9 (Viral title: $>1 \times 10^{13}$, Vector Biolab

AAV9-CMV7-sgRai1-GFP (Viral title: $>1 \times 10^9$), Applied Biological Materials Inc. Three sgRNAs were pooled to target Rai1 (sgRNA1: TTCCTCGCCAGAGTAGCGCC, sgRNA2: CCCAGCCTCATGATAGGCCG, sgRNA3: CCCAGCCTCATGATAGGCCG).

Immunostaining of mouse brain tissues

Immunostaining experiments were performed as previously described (Huang *et al.*, 2012 [\[1\]](#)). Briefly, mice were anesthetized and perfused with 1X PBS and 4% formaldehyde, and the brains were dissected and fixed in 4% formaldehyde overnight. Fixed brains were then transferred to 30% sucrose solution and subsequently embedded in the OCT solution. The brains were then sectioned (40 μ m), washed in PBS, permeabilized with 0.5% Triton X-100, and incubated with 10% normal donkey serum in PBS for 2 hours. The sections were then incubated overnight with primary antibodies (Rai1: 1:500 dilution, RRID:AB_2921229 (Huang *et al.*, 2016 [\[2\]](#)); Bdnf: 1:500, RRID:AB_10862052, Abcam ab108319; p-TrkB: 1:500 dilution, RRID:AB_2721199, Millipore ABN1381; oxytocin: 1:1000 dilution, RRID:AB_11212999, Millipore MAB5296) at 4°C on a shaker. The following day, sections were washed with 1X PBS twice for 10 minutes and incubated with a secondary antibody (1:1000 dilution) for 3 hours at room temperature. Sections were washed with 1X PBS once before mounting on glass slides with the DAPI-mounting (DAPI-Fluoromount-G, SouthernBiotech) and then imaged using a confocal microscope (Olympus FV-1000 confocal laser scanning microscope) with 20x or 40x objectives (NA 0.85 and 1.3, respectively). Z-stacks were taken with a step size of 1.0 μ m with a 1024 x 1024 resolution. Images were acquired as a stack with at least 13 optical sections. Image analysis was performed using the Fiji software.

Real-Time Quantitative Reverse Transcription PCR (qRT-PCR)

qRT-PCR was performed using mRNAs isolated from hypothalamic tissue samples. After isolation of total RNA, mRNAs were reverse-transcribed with the SuperScript III First-Strand Synthesis System (Thermo-Fisher). Quantitative PCR reactions were conducted using SsoFast EvaGreen Supermix on a Bio-Rad qPCR system.

Enzyme-linked immunosorbent assay (ELISA)

ELISA was performed using hypothalamic tissues from 7-week-old SMS mice to assess the mature Bdnf protein levels using a commercially available Bdnf ELISA kit (Mature BDNF Rapid ELISA Kit: Human, Mouse, Rat).

Western blotting analysis

Mice were quickly decapitated, brains were rapidly removed, and hypothalamic tissues were collected, weighed, and snap-frozen with liquid nitrogen. Hypothalamic tissues were lysed using a tissue lysis buffer containing RIPA buffer (5M NaCl, 1M PH 8.0 Tris HCl, 1% NP.40, 10% $C_{24}H_{39}NaO_4$, and 20% SDS) and the Halt™ Protease and Phosphatase Inhibitor Cocktail (Thermo-Fisher). Lysates were first sonicated on ice for 15 seconds followed by centrifuging at 10000 RPM at 4°C for 10 minutes. The supernatant was collected in a separate Eppendorf tube, and a BCA assay was performed to quantify protein levels. Proteins were then separated using SDS-PAGE gel and transferred onto nitrocellulose membranes. To determine the phosphorylation levels of Akt, membranes were cut and blocked in 5% milk in tris-buffered saline (TBS) for 1 hour and then incubated overnight at 4°C with anti-p-Akt (Ser473) antibody (1:1000 dilution, RRID:AB_2315049, Cell Signaling Technology 4060T) and anti-H3 antibody (internal control: 1:1000 dilution, RRID:AB_302613, Abcam ab1791). The next day, membranes were washed with TBST three times for five minutes before incubating with a HRP-conjugated secondary antibody (1:10000 dilution, Thermo Fisher 200-403-095S). Membranes were later stripped by incubating in 0.5M NaOH for 10 min and then re-probed for total Akt levels by incubating with anti-Akt (C67E7) antibody (1:1000 dilution, Cell Signaling Technology 4691T). The density of each immunoreactive phospho-protein band was expressed as a fraction of the band for the total Akt protein in the same lane.

Reverse-phase protein array (RPPA)

Reverse-phase protein array (RPPA) was used for precise high-throughput quantification of total and phospho-protein levels in different signalling pathways. Protein lysates from hypothalamic tissues were extracted and transferred to 384-well microarray plates to print individual slides. The slides were then labelled with validated antibodies that recognize total protein and specific phosphorylation sites. SYPRO Ruby staining was used to measure total protein levels. The data obtained from RPPA were normalized for individual samples. Each antibody's raw signal intensity (SI) was determined by subtracting the background from the antibody-specific signals during image analysis. The normalized antibody SI was expressed using the formula $N = (A - C) / T \times M$, where N is the normalized SI, A is the raw antibody SI, C is the negative control SI, T is the SI of total protein, and M is the median SI of total protein from a spot within the same sample group. Antibodies with SI less than 200 were filtered out, and quality control (QC) scores were established to discard poor-quality reactions. The coefficient of variation (CV) score was determined for each antibody and defined as the percentage of samples on a slide with a CV < 20%. Antibodies with a median technical triplicate CV > 20% were also excluded from further analysis. T-tests were performed for each antibody to identify differentially expressed proteins in the SMS compared to control groups. PCA plots were generated for all samples, annotated with their experimental group, and hierarchical clustering was performed using a Python analysis tool. Enrichment analysis on the differentially expressed proteins was performed using an online data resource string: version 11.5.

Electrophysiology of acute brain slices

Mouse brains were removed and placed in ice-cold carbonated slicing artificial cerebrospinal fluid (aCSF), which contained (in mM) 124 NaCl, 3 KCl, 1.4 NaH_2PO_4 , 26 $NaHCO_3$, 1.3 Mg SO_4 , 10 D-glucose, 0.5 $CaCl_2$, and 3.5 $MgCl_2$. Coronal sections (300 μm) were cut on a vibratome. Slices were allowed to recover at 32 °C for 40 minutes and then at 30 °C for 30 minutes to 6 hours in the recording medium, carbonated aCSF (124 NaCl, 3 KCl, 1.4 NaH_2PO_4 , 26 $NaHCO_3$, 1.3 Mg SO_4 , 10 D-glucose, 2.4 $CaCl_2$, in mM, ~300 mOsm). Signals were recorded with a 5× gain, low-pass filtered at 2

kHz, digitized at 10 kHz (Molecular Devices Multiclamp 700B) and analyzed with pClamp 11 (Molecular Devices). Whole-cell recordings were made using 3–5 M Ω pipettes filled with an internal solution that contained (in mM) 131 potassium gluconate, 8 NaCl, 20 KCl, 2 EGTA, 10 HEPES, 2 MgATP and 0.3 Na3GTP (pH 7.2, with KOH, 300–310 mOsm with sucrose). For miniature current recording, the internal solution contained (in mM) 130 gluconate acid, 8 CsCl, 1 NaCl, 2 EGTA, 0.8 CsOH, 10 HEPES, 2 MgATP and 0.3 Na3GTP (pH 7.2, with CsOH, 300–310 mOsm adjusted with sucrose). mEPSCs were recorded in the presence of 1 μ M TTX at a holding potential of -70 mV. mIPSCs were recorded at a holding potential of +10 mV. The mIPSCs were analyzed from events during a 2-minute continuous recording that contained > 15 events/neuron. Additional calculations, statistical analysis, and plotting was performed in Microsoft Excel or Prism (GraphPad).

Spectrometric analysis of LM22A-4 levels in the brain

We assessed the efficacy of LM22A-4, a small molecule pharmacological agent, in 8-week-old SMS and control mice. LM22A-4, chemical name N1, N3, N5tris (2hydroxyethyl) 1, 3, 5 benzenetricarboxamide (Molecular formula: C₁₅H₂₁N₃O₆, Formula weight: 339.3), was manufactured by Cayman Chemical with purity specification >98%. A stock solution was made by first dissolving LM22A-4 in DMSO (solubility =30mg/ml) and then diluting it with 0.9% isotonic sterilized saline solution (Sigma). To maximize brain concentrations, LM22A-4 was administered both intranasally (5mg/kg) and intraperitoneally (50mg/kg), as previously reported (Simmons *et al.*, 2013 [\[1\]](#)). The bioavailability of this drug in the hypothalamus using these doses and routes of administration was assessed with the reverse phase LC triple-quadrupole tandem mass spectroscopic detection (LC-MS/MS) by Absorption Systems. Mice were euthanized, and hypothalamic tissue was collected at either 1 h or 3 h post-injection (n= 4/timepoint). Known amounts of LM22A-4 were added to the blank sample to generate a standard curve to verify test accuracy.

LM22A-4 treatment

Mice were divided into four groups: Saline-injected control, Saline-injected SMS, LM22A-4-injected SMS, and LM22A-4-injected control mice. Mice were injected daily, intra-nasally (5mg/kg) and intraperitoneally (50mg/kg), for 7 weeks. Body weight was measured between 8 and 14 weeks of age. Food intake was measured from 14 to 15 weeks of age. After 15 weeks of age, mice were euthanized to collect brown adipose, visceral, subcutaneous (inguinal) and ependymal white adipose tissues and weighed using an analytical scale (Sartorius).

Hypothalamic brain tissues were collected for western blot analysis. Metabolic profiling was performed on a separate cohort of mice injected for two weeks prior to the start of experiments as well as at specific weeks during the experiment.

Vertical rearing

To measure vertical repetitive rearing, mice were taken into the testing room and allowed to habituate for 1 h prior to testing. During the test, mice were placed in the center of a 45 cm (W) $\times$ 45 cm (D) $\times$ 40 cm (H) square arena, and their movement was recorded for 10 minutes.

Statistical analysis

Sample sizes used were based on our previously studies (Huang *et al.*, 2016 [\[2\]](#); Javed *et al.*, 2021 [\[3\]](#)). The data were subjected to statistical analysis for significance by using GraphPad Prism 9.9. Significance was defined as having a p-value < 0.05. The significance levels were expressed as follows: * p< 0.05, **p<0.01, ***p<0.0001, **** p<0.0001. The statistical tests used for each analysis (e.g., Student's t-test, Fisher's exact test, analysis of variance [ANOVA]) are indicated in the text and legends.

Data Availability

This study includes no data deposited in external repositories.

Acknowledgements

Sehrish Javed was supported by a doctoral award from the Fonds de Recherche du Québec - Santé (FRQS). This work was supported by the Rare Diseases: Models & Mechanisms Network and Smith-Magenis syndrome Research Foundation.

Author contributions

Sehrish Javed: Conceptualization; formal analysis; investigation; data curation; visualization; writing. **Ya-Ting Chang:** Formal analysis; investigation; data curation; visualization. **Yoobin Cho:** Formal analysis; investigation. **Yu-Ju Lee:** Formal analysis; investigation. **Hao-Cheng Chang:** Formal analysis; investigation. **Minza Haque:** Formal analysis; investigation. **Yu Cheng Lin:** Formal analysis; investigation. **Wei-Hsiang Huang:** Conceptualization; supervision; funding acquisition; visualization; writing; project administration.

Disclosure and competing interests statement

The authors do not have any conflicts of interest to disclose relating to this work.

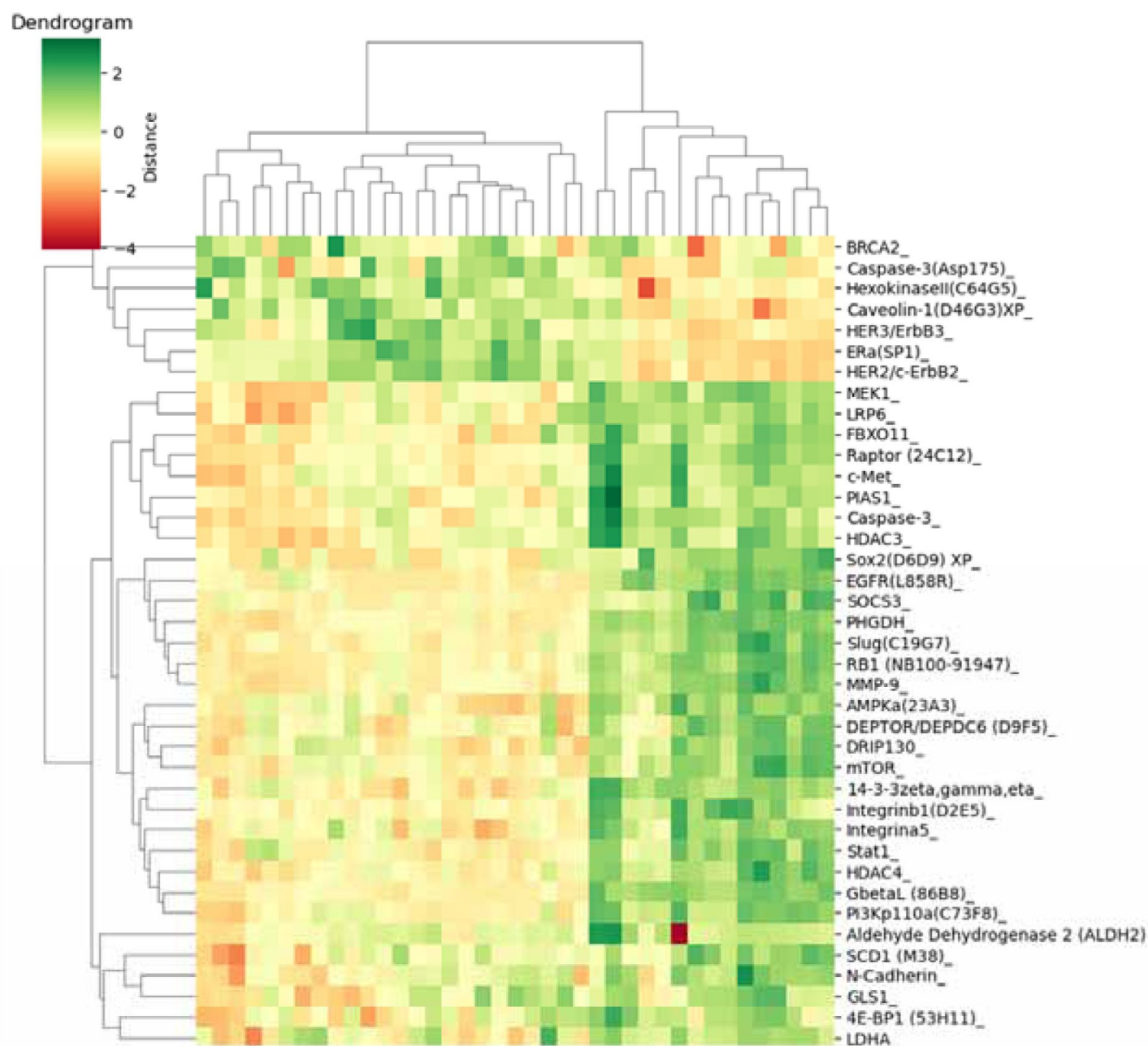


Fig 1-figure supplement 1

Differentially expressed proteins in the hypothalamus of SMS mice

A heat map showing hierarchical clustering of differentially expressed proteins in the hypothalamus of SMS mice.

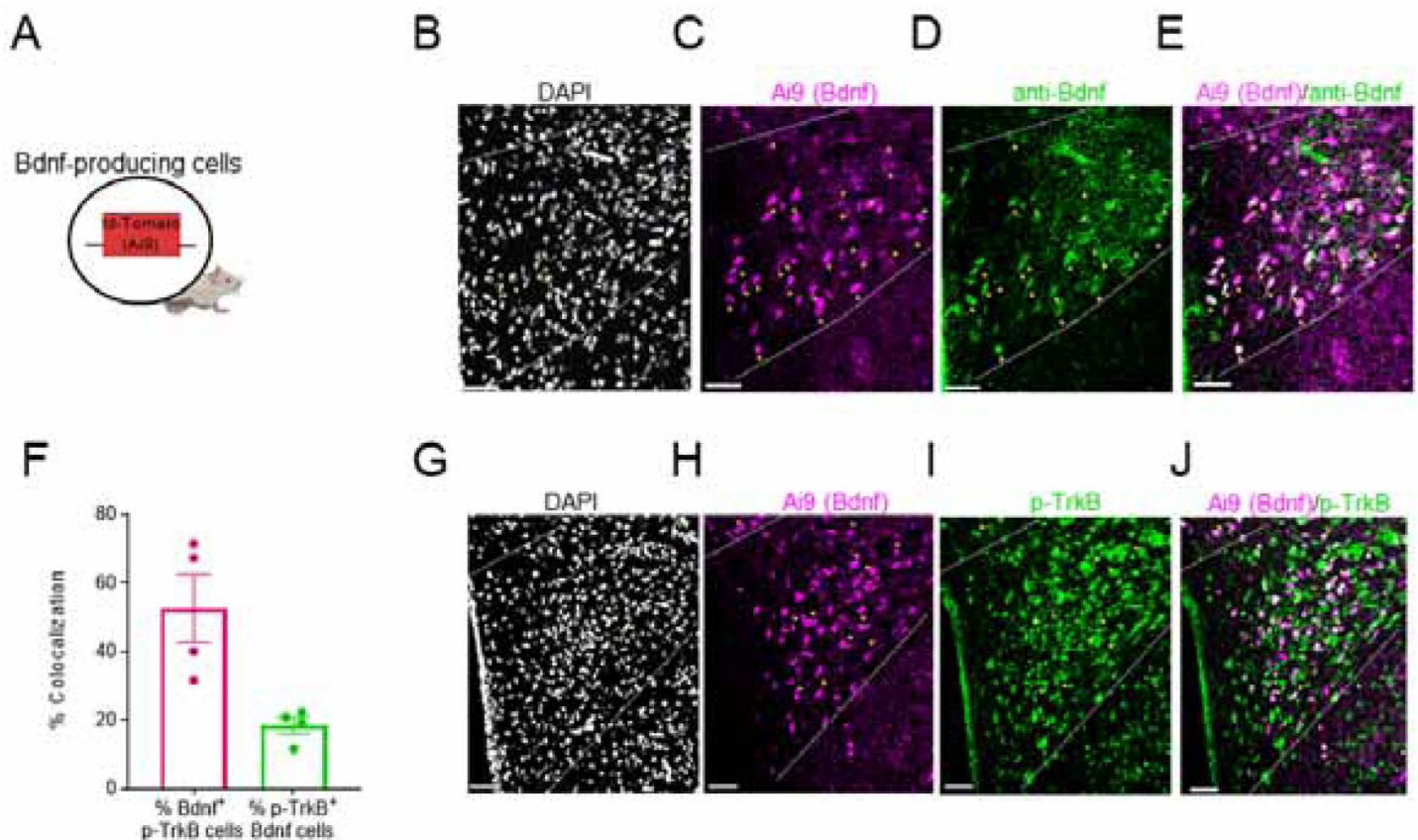


Fig 2-figure supplement 1

The *Bdnf^{Cre}* allele labels BDNF-producing neurons in the paraventricular nucleus of hypothalamus

(A) Schematic showing a transgenic mouse expressing *Bdnf^{Cre}*-dependent td-Tomato (Ai9) signals in BDNF-expressing cells.

(B-E) Colocalization of *Bdnf^{Cre}*-dependent td-Tomato signals with endogenous BDNF protein. Black: DAPI, Magenta: BDNF-producing cells, Green: endogenous BDNF. Scale bar: 50 μ m.

(F) Quantification showing the percentage of BDNF-producing cells that express p-TrkB protein (left bar, magenta) and the percentage of p-TrkB-expressing cells that express BDNF (right bar, green) in the PVH. Data are shown as mean $\pm$ SEM.

(H-J) Representative images showing that BDNF-producing cells and p-TrkB signals are colocalized in the PVH. Black: DAPI, Magenta: BDNF-producing cells, Green: phospho-TrkB⁸¹⁶ (Scale bar: 50 μ m).

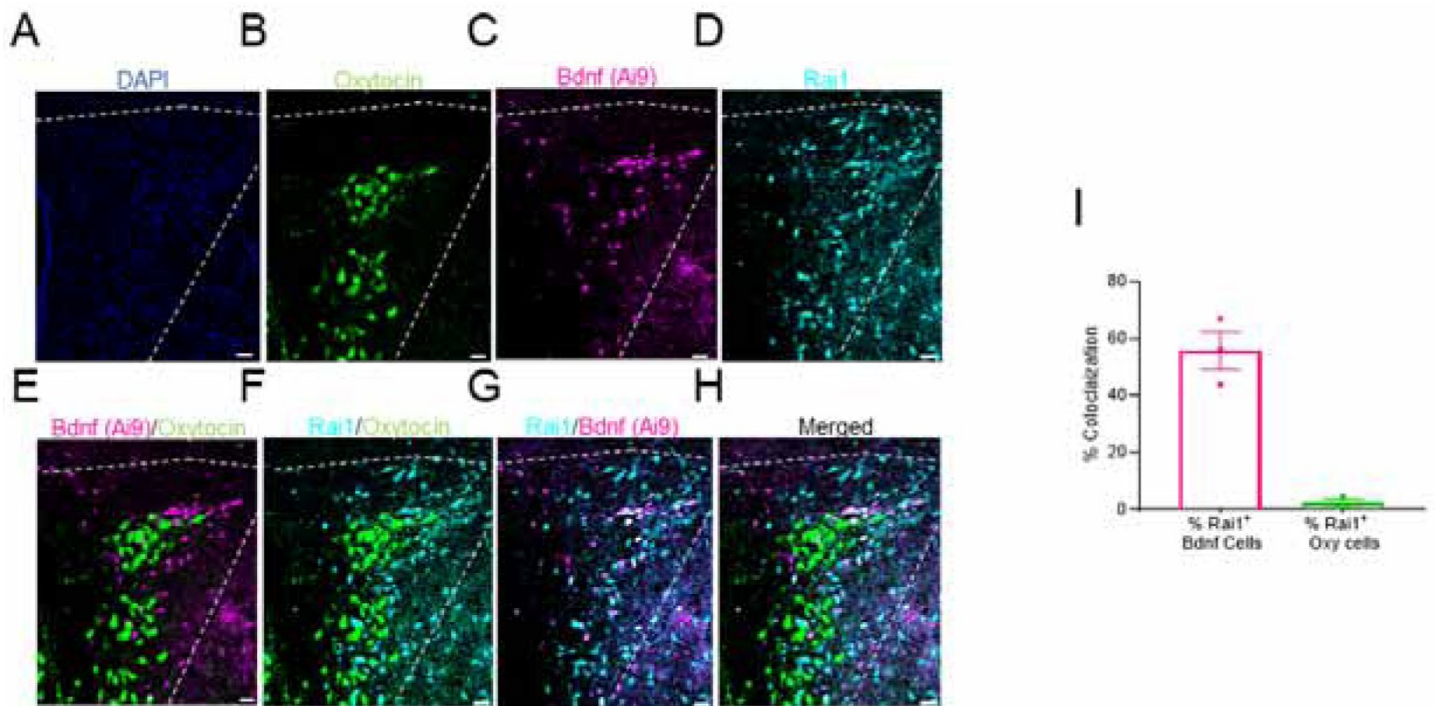


Fig 2-figure supplement 2

Rai1 expression in Bdnf-expressing but not oxytocin-expressing magnocellular PVH neurons

(A-H) Representative images DAPI expression (blue, A), oxytocin (green, B), Bdnf-producing cells (Ai9 reporter, magenta, C), and Rai1 protein (cyan, D) in the PVH. Note that Bdnf-producing neurons are distinct from the oxytocin-expressing neurons (E) and that Rai1 is more selectively expressed in the Bdnf-expressing neurons (F-H).

(I) Quantification showing the percentage of Rai1⁺ cells that co-express either Bdnf (left bar, magenta) or oxytocin (right bar, green) in the PVH. n=3 mice/group. Data are shown as mean±SEM.

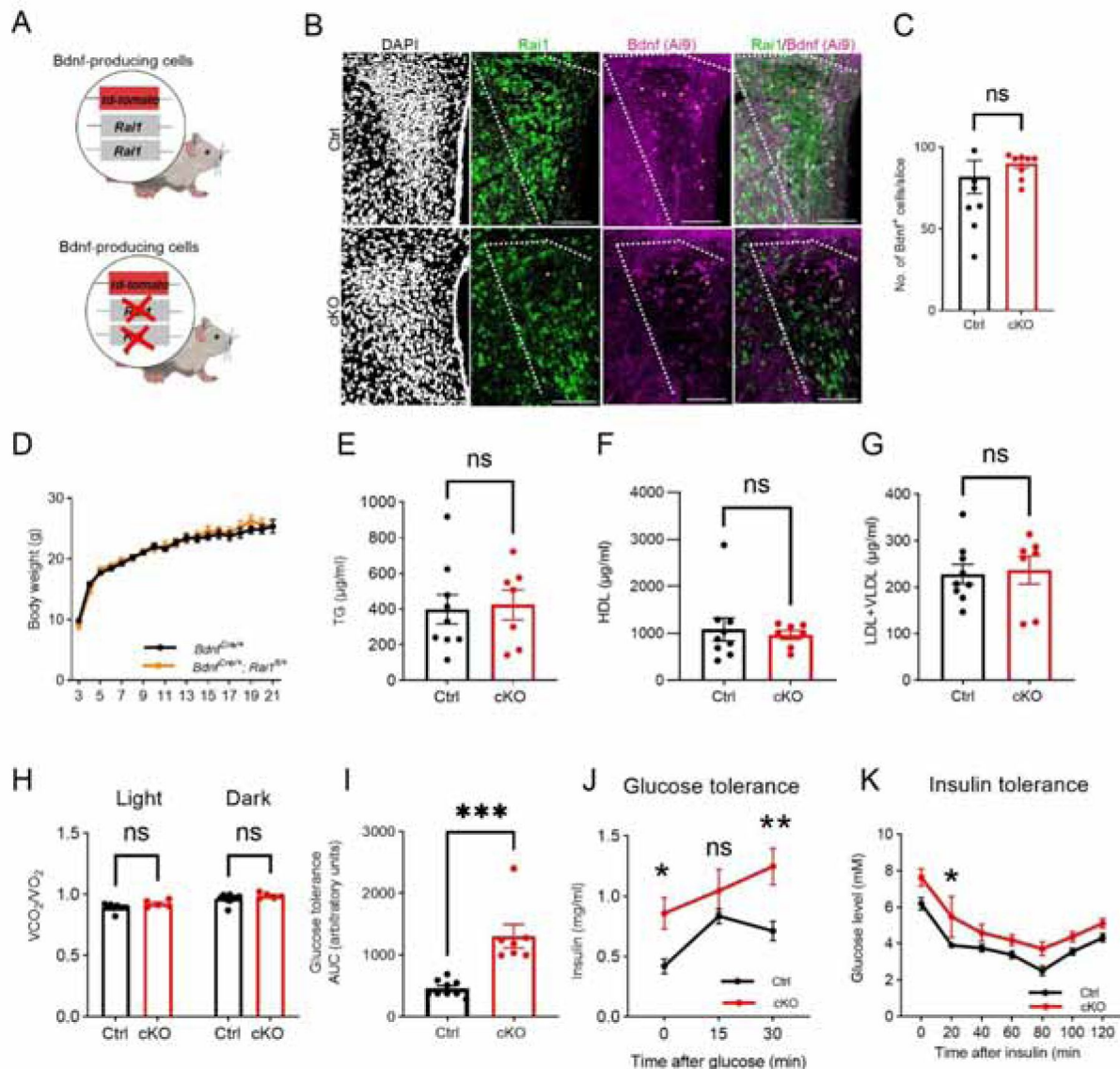


Fig 2-figure supplement 3

Ablating *Rai1* from the *Bdnf*-producing cells induces body weight gain and defective energy homeostasis

(A) Schematic representation of the conditional knockout mice carrying either normal *Rai1* alleles and the Ai9 reporter in *Bdnf*-producing cells (top) or homozygous *Rai1* deletion and the Ai9 reporter in the *Bdnf*-producing cells.

(B) Representative images showing the expression of DAPI (white), *Rai1*-expressing cells (green), and *Bdnf*-expressing cells (Ai9 reporter, magenta) in the PVH. The top panel shows the Ctrl group with *Rai1*-Ai9 co-expression (yellow arrowheads) and the bottom panel shows reduced *Rai1* and Ai9 colocalization (yellow arrowheads). Note that *Rai1* expression was still detected in non-*Bdnf* neurons in the PVH.

(C) Quantification showing the percentage of *Bdnf*-producing cells was not altered by *Rai1* deletion, indicating that *Rai1* loss did not impair the generation or survival of PVH^{*Bdnf*} neurons. ns indicates not significantly different, student's t-test.

(D) The *Bdnf*^{Cre/+} and *Bdnf*^{Cre/+}; *Rai1*^{fl/fl} female mice had similar body weights.

(E-G) Blood parameter analysis of triglycerides (TG), high-density lipoprotein (HDL), low-density lipoprotein and very low-density lipoprotein (LDL+VLDL) did not differ between Ctrl and cKO mice. ns indicates not significantly different, student's t-test.

(H) Respiratory exchange rate did not differ between Ctrl and cKO mice.

(I) In the glucose tolerance test, the cKO mice showed an increased area under curve (AUC). ****p* < 0.001, student's t-test.

(J) In the glucose tolerance test, cKO mice showed increased insulin levels at the beginning of the test and at 30 minutes after glucose administration.

(K) In the insulin tolerance test, blood glucose level after insulin injection.

Data are shown as mean ± SEM. Statistics for D, H, J-K: ns indicates not significantly different, **p* < 0.05, ***p* < 0.01, ****p* < 0.001, *****p* < 0.0001, two-way ANOVA with Šidák's multiple comparisons test.

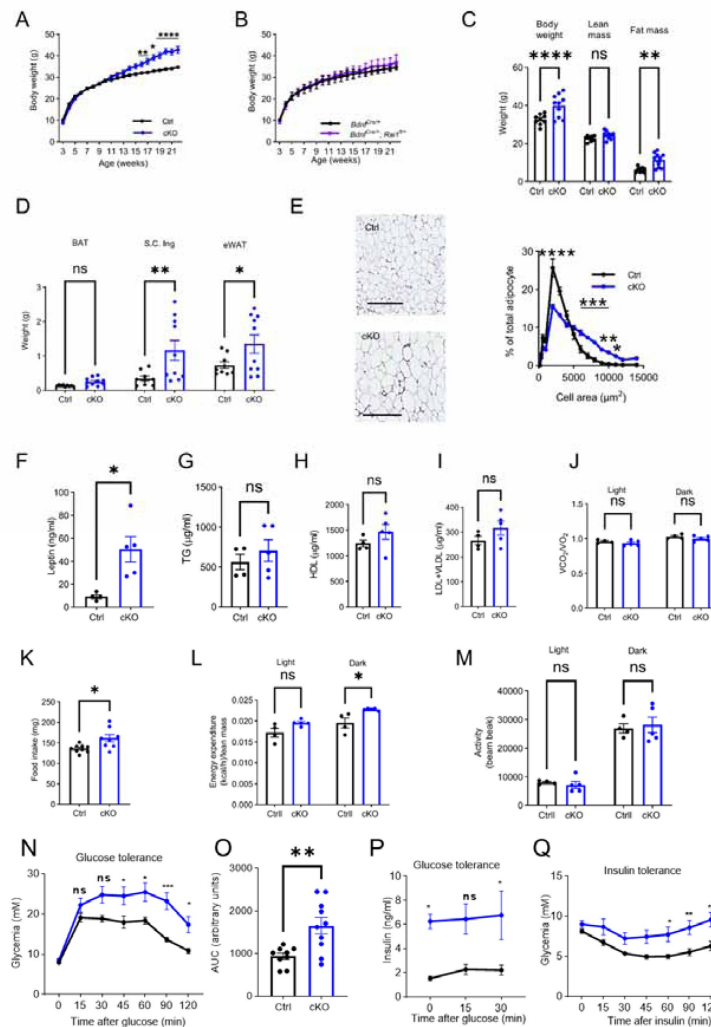


Fig 2-figure supplement 4

Metabolic profiles of male cKO mice lacking *Rai1* expression in the *Bdnf*-producing cells

(A) Male cKO mice gained significantly more weight than Ctrl mice beginning at 15 weeks of age.

(B) Male *Bdnf^{Cre/+}* and *Bdnf^{Cre/+}; Rai1^{fl/+}* mice had similar body weights.

(C) Echo-MRI analysis showed that 26-week-old male cKO mice had a significantly increased body weight due to increased fat but not lean mass. ****p<0.0001, student's t-test.

(D) Fat disposition analysis showing the weight of brown adipocytes (BAT), subcutaneous inguinal (S.C.ing), and epididymal white adipose tissues (eWAT). ns indicates not significantly different, *p<0.05, **p<0.01, student's t-test.

(E) Male cKO mice showed eWAT cell hypertrophy. Top: Representative images of the eWAT tissues in Ctrl and cKO mice. Scale bar = 500μm. Bottom: Frequency distribution of cellular sizes.

(F-I) Blood parameter analysis showing that male cKO mice showed significantly increased leptin levels (F) without alterations in TG (G), HDL (H), LDL+VLDL (I). *p<0.05, student's t-test.

(J) No differences were found in respiratory exchange rate in male Ctrl and cKO mice.

(K) Male cKO mice showed a significantly increased food intake compared to Ctrl littermates. *p<0.05, student's t-test.

(L) Male cKO mice showed increased energy expenditure at the dark phase.

(M) Male Ctrl and cKO mice showed similar locomotor activities.

(N-P) Glucose tolerance test shows that cKO male mice are glycemic 45 minutes post intraperitoneal glucose administration (N), increased AUC (O), despite significantly higher insulin levels before and 30 minutes after glucose administration (P), suggesting potential insulin insensitivity.

(Q) Insulin tolerance test showing that male cKO mice showed significantly higher blood glucose levels 60 minutes after intraperitoneal insulin administration. **p<0.01, student's t-test.

Data are shown as mean±SEM. Statistics for A-B, E, J, L-N, P-Q: ns indicates not significantly different, *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001, one- or two-way ANOVA with Šidák's multiple comparisons test.

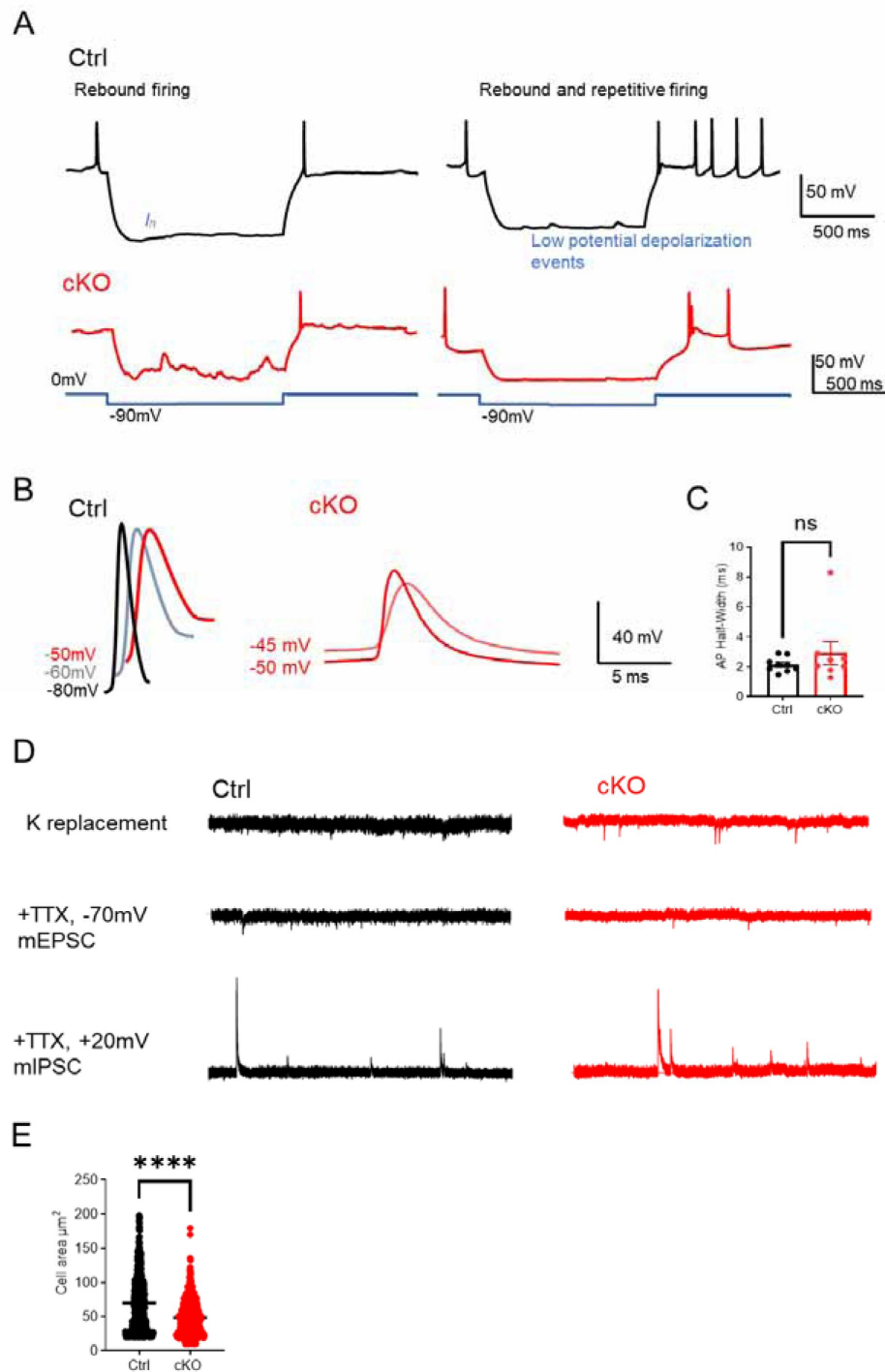


Fig 3-figure supplement 1

The cellular and synaptic properties of control and Rai1-deficient PVH^{Bdnf} neurons

- (A) Representative traces of rebound and rebound repetitive firing of PVH^{Bdnf} neurons in control (black) and cKO (red).
- (B) Representative AP waveforms at holding voltages of -70mV, -60mV, and -50mV in control (left) and -50mV and -45mV in cKO (right).
- (C) The half-width of AP initiated at a holding voltage of -60mV (neuronal numbers in brackets, p=0.374, student's t-test).
- (D) Representative traces of miniature recordings in control (black) and cKO (red) PVH^{Bdnf} neurons. The traces were recorded with a Cs-based internal at a holding voltage of -70mV in aCSF (top), spontaneous mEPSCs were recorded in aCSF containing 1μM TTX at a holding voltage of -70mV (middle), and mIPSCs were recorded at holding voltage of +10mV (bottom).
- (E) Quantification showing that the cellular size of cKO PVH^{Bdnf} neurons was significantly smaller than Ctrl neurons. ****p<0.0001, student's t-test.

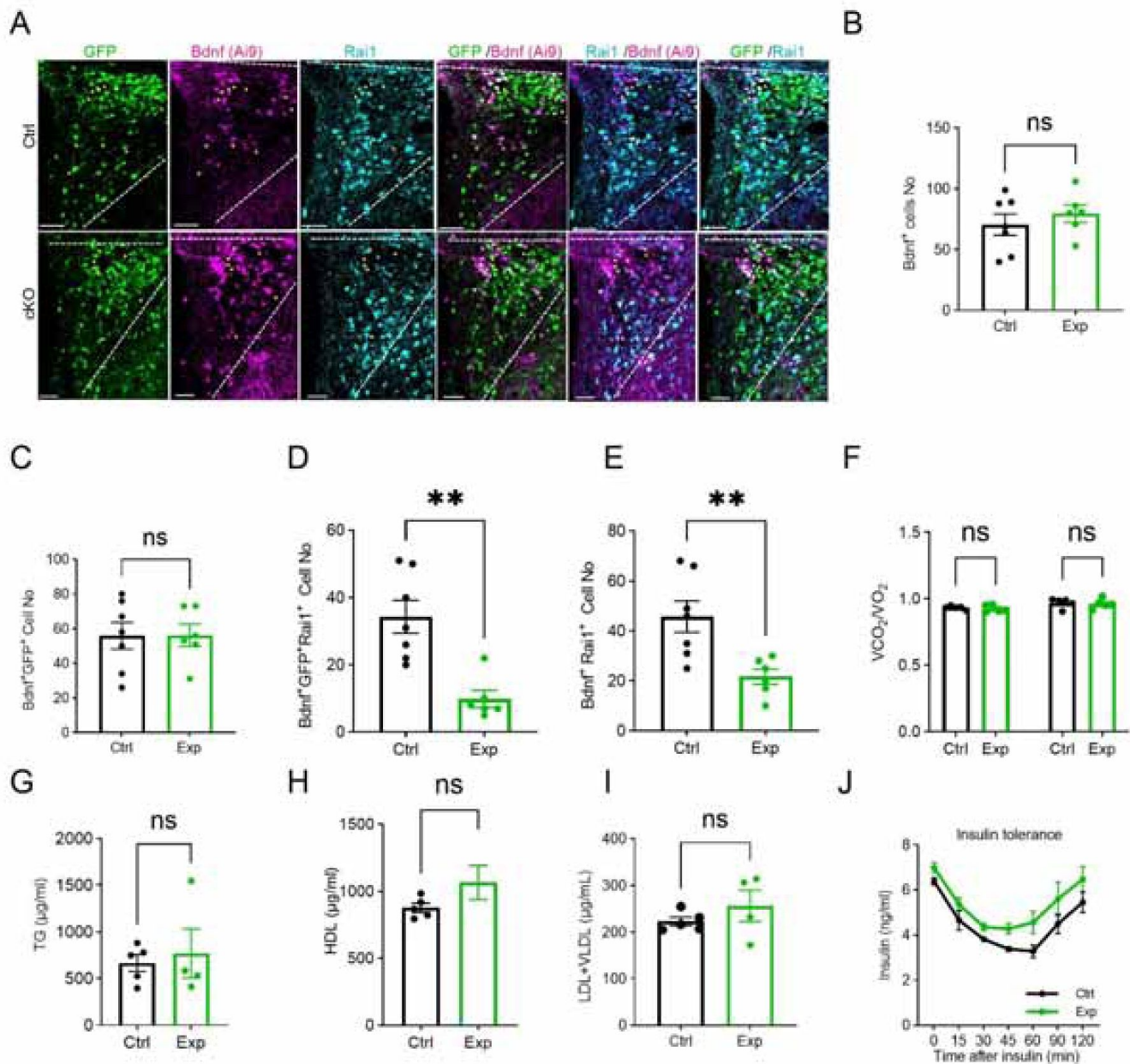


Fig 4-figure supplement 1

Metabolic profile of mice lacking *Rai1* in PVH^{Bdnf} neurons

(A) Representative images showing co-expression of virus-infected cells (GFP/green), *Bdnf*^{Cre}-dependent Ai9 signals (Bdnf cells, magenta), and endogenous Rai1 (cyan). Yellow arrowheads indicate Bdnf neurons infected with AAV and in these cells, Rai1 signals were diminished in Exp but not Ctrl mice. Scale bar = 50μm.

(B-C) The total number of PVH^{Bdnf} neurons **(B)** and total number of PVH^{Bdnf} neurons infected with AAVs **(C)** per slice showing that Rai1 deletion or virus infection did not alter the total number of PVH^{Bdnf} neurons.

(D) Rai1 expression was significantly reduced in PVH^{Bdnf} neurons infected with both AAVs when compared to PVH^{Bdnf} neurons infected with sgRai1-GFP AAV alone.

(E) The number of PVH^{Bdnf} neurons that co-express Rai1 was significantly reduced in the Exp group. **(F)** Respiratory exchange rate remained similar in Ctrl and Exp mice.

(G-I) Blood parameter analysis showing the levels of TG, HDL, and LDL+VLDL were similar in Ctrl and Exp mice.

(J) In the insulin tolerance test, Exp group showed a trend towards increased blood glucose level after insulin injection.

Data are shown as mean±SEM. Statistics for **F, J**: ns indicates not significantly different, two-way ANOVA. Statistics for **B-E, G-I**: **p<0.01, student's t-test.

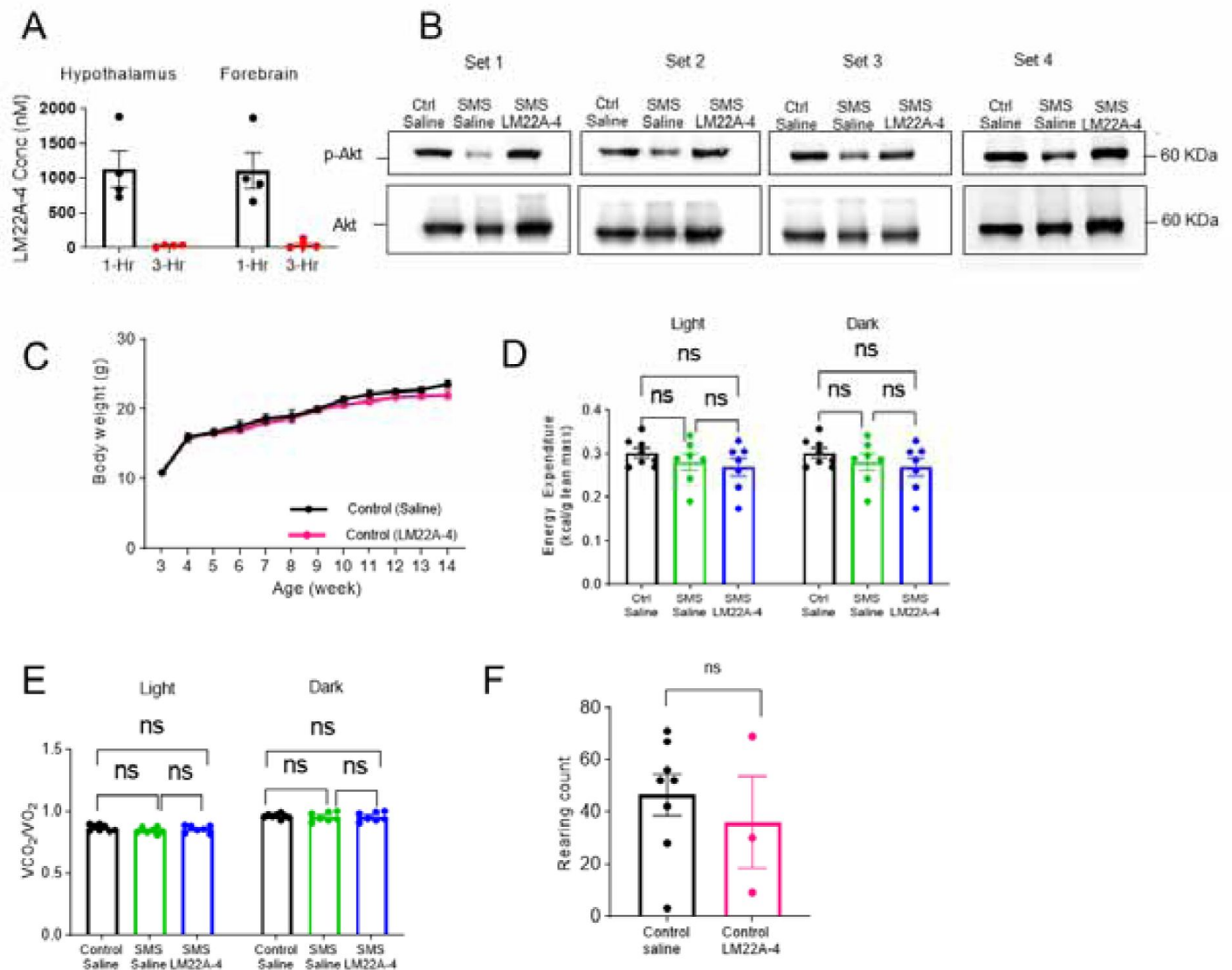


Figure 5—figure supplement 1

LM22A-4 treatment does not disrupt energy homeostasis and repetitive rearing in the Ctrl mice

(A) LM22A-4 concentration in the hypothalamus and forebrain of SMS mice 1 hour (black dots) and 3 hours (red dots) after simultaneous intranasal and IP injections.

(B) Four independent sets of western blotting analyses showing reduced p-Akt levels in the hypothalamus of saline-treated SMS mice, which were increased by LM22A-4 treatment. Total Akt protein was used as a control.

(C) Similar body weight of Ctrl mice treated with either saline or LM22A-4.

(D) Energy expenditure was not different between groups in the light vs. dark phase.

(E) Respiratory exchange rates were not different between groups in the light vs. dark phase.

(F) Repetitive rearing behaviour in Ctrl mice was not altered by LM22A-4 treatment. ns indicates not significantly different, student's t-test.

Data are shown as mean±SEM. Statistics for **C-E**: ns indicates not significantly different, two-way ANOVA. Statistics for **F**: student's t-test.

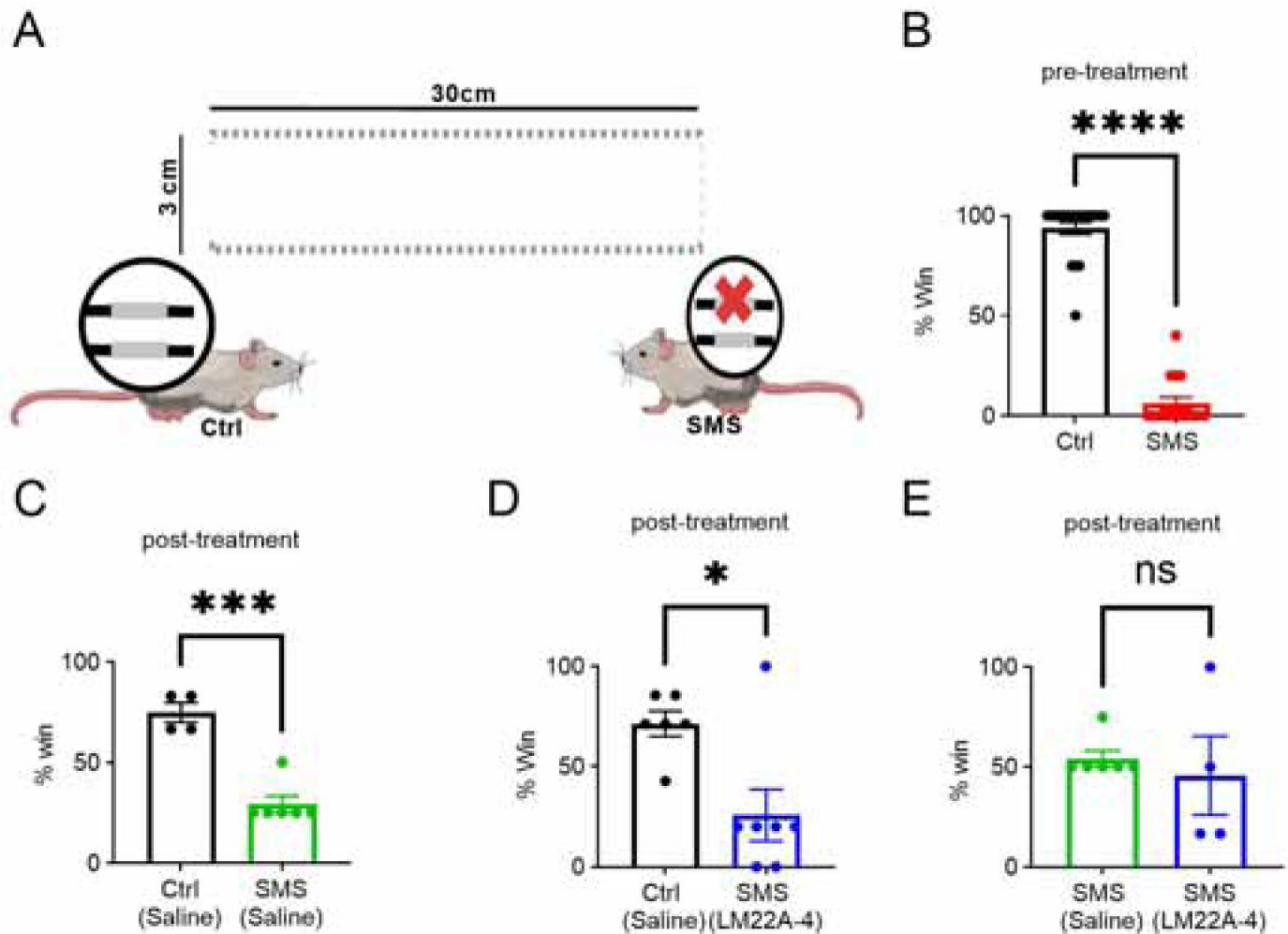


Figure 5—figure supplement 2

LM22A-4 treatment in adult SMS mice is insufficient to improve social interaction deficit

- (A) Schematic showing the tube test that assesses social interaction between stranger mice of different genotypes.
- (B) 7-week-old Ctrl mice won significantly more than SMS mice before the LM22A-4 treatment.
- (C) 10-week-old saline-injected Ctrl mice won significantly more than the saline-injected SMS mice (two weeks post-LM22A-4 treatment).
- (D) 10-week-old saline-injected Ctrl mice won significantly more than LM22A-4 injected SMS mice.
- (E) 10-week-old saline-injected Ctrl mice showed a similar winning rate to LM22A-4-injected SMS mice.
- Data are shown as mean±SEM. Statistics for **B-E**, ns indicates not significantly different, * $p < 0.05$, *** $p < 0.001$, **** $p < 0.0001$, student's t-test.

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Reviewer #1 (Public Review):

Summary:

Rai1 encodes the transcription factor retinoic acid-induced 1 (RAI1), which regulates expression of factors involved in neuronal development and synaptic transmission. Rai1 haploinsufficiency leads to the monogenic disorder Smith-Magenis syndrome (SMS), which is associated with excessive feeding, obesity and intellectual disability. Consistent with findings in human subjects, Rai1^{+/-} mice and mice with conditional deletion of Rai1 in Sim⁺ neurons, which are abundant in the paraventricular nucleus (PVN), exhibit hyperphagia, obesity and

increased adiposity. Furthermore, RAI1-deficient mice exhibit reduced expression of brain-derived neurotrophic factor (BDNF), a satiety factor essential for the central control of energy balance. Notably, overexpression of BDNF in PVN of RAI1-deficient mice mitigated their obesity, implicating this neurotrophin in the metabolic dysfunction these animals exhibit. In this follow up study, Javed et al. interrogated the necessity of RAI1 in BDNF+ neurons promoting metabolic health.

Consistent with previous reports, the authors observed reduced BDNF expression in the hypothalamus of *Rai1*^{+/-} mice. Moreover, proteomics analysis indicated impairment in neurotrophin signaling in the mutants. Selective deletion of *Rai1* in BDNF+ neurons in the brain during development resulted in increased body weight, fat mass and reduced locomotor activity and energy expenditure without changes in food intake. There was also a robust effect on glycemic control, with mutants exhibiting glucose intolerance. Selective depletion of RAI1 in BDNF+ neurons in PVN in adult mice also resulted in increased body weight, reduced locomotor activity, and glucose intolerance without affecting food intake. Blunting RAI1 activity also leads to increases and decreases in the inhibitory tone and intrinsic excitability, respectively, of BDNF+ neurons in the PVN.

Strengths:

Overall, the experiments are well designed and multidisciplinary approaches are employed to demonstrate that RAI1 deficits in BDNF+ neurons diminish hypothalamic BDNF signaling and produce metabolic dysfunction. The most significant advance relative to previous reports is the finding from electrophysiological studies showing that blunting RAI1 activity leads to increases and decreases the inhibitory tone and intrinsic excitability, respectively, of BDNF+ neurons in the PVN. Furthermore, that intact RAI1 function is required in BDNF+ neurons for the regulation of glucose homeostasis.

Weaknesses:

Some of the data need to be reconciled with previous findings by others. For example, the authors report that more than 50% of BDNF+ neurons in PVN also express pTrkB whereas about 20% of pTrkB+ cells contain BDNF, raising the possibility that autocrine mechanisms might be at play. This is in conflict with a previous study by An et al, (2015) showing that these cell populations are largely non-overlapping in the PVN.

Another issue that deserves more in depth discussion is that diminished BDNF function appears to play a minor part driving deficits in energy balance regulation. Accordingly, both global central depletion of *Rai1* in BDNF+ neurons during development and deletion of *Rai1* in BDNF+ neurons in the adult PVN elicited modest effects on body weight (less than 18% increase) and did not affect food intake. This contrasts with mice with selective *Bdnf* deletion in the adult PVN, which are hyperphagic and dramatically obese (90% heavier than controls). Therefore, the results suggest that deficits in RAI1 in PVN or the whole brain only moderately affect BDNF actions influencing energy homeostasis and that other signaling cascades and neuronal populations play a more prominent role driving the phenotypes observed in *Rai1*^{+/-} mice, which are hyperphagic and 95% heavier than controls. The results from the proteomic analysis of hypothalamic tissue of *Rai1* mutant mice and controls could be useful in generating alternative hypotheses.

Depleting RAI1 in BDNF+ neurons had a robust effect compromising glycemic control. However, as the approach does not necessarily impact BDNF exclusively, there should be a larger discussion of alternative mechanisms.

Reviewer #2 (Public Review):

Understanding disease conditions often yields valuable insights into the physiological regulation of biological functions, as well as potential therapeutic approaches. In previous investigations, the author's research group identified abnormal expression of brain-derived

neurotrophic factor (BDNF) in the hypothalamus of a mouse model exhibiting Smith-Magenis syndrome (SMS), which is caused by heterozygous mutations of the *Rai1* gene. Human SMS is associated with distinct facial characteristics, sleep disturbances, behavioral issues, and intellectual disabilities, often accompanied by obesity. Conditional knockout (cKO) of the *Bdnf* gene from the paraventricular hypothalamus (PVH) in mice led to hyperphagic obesity, while overexpression of the *Bdnf* gene in the PVH of *Rai1* heterozygous mice restored the SMS-like obese phenotype. Based on these preceding findings, the authors of the present study discovered that homozygous *Rai1* cKO restricted to *Bdnf*-expressing cells, or *Rai1* gene knockdown solely in *Bdnf*-positive neurons in the PVH, induced obesity along with intricate alterations in adipose tissue composition, energy expenditure, locomotion, feeding patterns, and glucose tolerance, some of which varied between sexes. Additionally, the authors demonstrated that a brain-penetrating drug capable of activating the TrkB pathway, a downstream signaling pathway of BDNF, partially alleviated the SMS-like obesity phenotype in female mice with *Rai1* heterozygous mutations. Although the specific (neural) cell type responsible for this TrkB signaling remains an open question, the present study unequivocally highlights the importance of *Rai1* gene function in PVH *Bdnf* neurons for the obesity phenotype, providing valuable insights into potential therapeutic strategies for managing obesity associated with SMS.

In the proteomic analysis (Fig. 1), the authors elucidated that multiple phospho-protein signaling pathways, including Akt and mTOR pathways, exhibited significant attenuation in the SMS model mice. Of significance, the manifestation of haploinsufficiency of the *Rai1* gene exclusively within the BDNF+ cells demonstrated negligible impact on body weight (Fig. 2-supple 3D), despite observing a reduction in BDNF levels in the heterozygous *Rai1* mutant (Fig. 1A). Conversely, the homozygous *Rai1* cKO in the BDNF+ cells prominently displayed an obesity phenotype, suggesting substantial dissimilarities in the gene expression profiles between *Rai1* heterozygous and homozygous conditions within the BDNF+ cell population. It would be advantageous to precisely identify the responsible differentially expressed genes, possibly including *Bdnf* itself, in the homozygous cKO model. The observed reduction in the excitability of PVH BDNF+ cells (Fig. 3) is presumably attributed to aberrant gene expression other than *Bdnf* itself, which may serve as a prospective target for gene expression analysis. Notably, the *Rai1* homozygous cKO mice in BDNF+ cells exhibited some sexual dimorphisms in feeding and energy expenditures, as evidenced by Fig. 2 and related figures. Exploring the potential relevance of these sexual differences to human SMS cases and investigating the underlying cellular/molecular mechanisms in the future would provide valuable insights.

Although the CRISPR-mediated knockdown of the *Rai1* gene (Fig. 4) appears to be highly effective, given the broad transduction of AAV serotype 9, it may be helpful to exclude the possibility of other brain regions adjacent to the PVH, such as the DMH or VMH, being affected by this viral procedure. If the PVH-specificity is established, the majority of *Rai1* cKO effects in *Bdnf*+ cells are primarily attributed to PVH-*Bdnf*+ cells based on the similarity of phenotypes observed. With regards to the apparent rescue of the body weight phenotype in *Rai1* heterozygous mutants using a selective TrkB activator, the specific biological processes, and neurons responsible for this effect remain unclear to this reviewer. Elucidating these aspects would be significant when considering potential applications to human SMS cases.

Overall, the present study represents a valuable addition to the authors' series of high-quality molecular genetic investigations into the *in vivo* functions of the *Rai1* gene. This reviewer particularly commends their diligent efforts to enhance our comprehension of SMS and contribute to the future development of more effective therapies for this syndrome.

Reviewer #3 (Public Review):

Summary:

Smith-Magenis syndrome (SMS) is associated with obesity and is caused by deletion or

mutations in one copy of the *Rai1* gene which encodes a transcriptional regulator. Previous studies have shown that *Bdnf* gene expression is reduced in the hypothalamus of *Rai1* heterozygous mice. This manuscript by Javed et al. further links SMS-associated obesity with reduced *Bdnf* gene expression in the PVH.

Strengths:

The authors show that deletion of the *Rai1* gene in all BDNF-expressing cells or just in the PVH BDNF neurons postnatally caused obesity. Interestingly, mutant mice displayed sexual dimorphism in the cause for the obesity phenotype. Overall, the data are well presented and convincing except the data from LM22A-4.

Weaknesses:

1. The most serious concern is about data from LM22A-4 administration experiments (Figure 5 and associated supplemental figures). A rigorous study has demonstrated that LM22A-4 does not activate TrkB (Boltaev et al., Science Signaling, 2017), which is consistent with unpublished results from many labs in the neurotrophin field. It is tricky to interpret body weight data from pharmacological studies because compounds always have some side effects, which can reduce body weight non-specifically.
2. The resolution of all figures are poor, and thus I could not judge the quality of the micrographs.
3. Citation of the literature is not precise. The study by An et al. (2015) shows that deletion of the *Bdnf* gene in the PVH leads to obesity due to increased food intake and reduced energy expenditure (not just hyperphagic obesity; Line 72). Furthermore, the study by Unger et al. (2017) carried out *Bdnf* deletion in the VMH and DMH using AAV-Cre and did not discuss SF1 neurons at all (Line 354). The two studies by Yang et al. (Mol Endocrinol, 2016) and Kamitakahara et al. (Mol Metab, 2015) did use SF1-Cre to delete the *Bdnf* gene and did not observe any obesity phenotype.
4. Animal number is not described in many figure legends.

Author Response

We express our sincere gratitude to the editors and reviewers for their invaluable input. To further improve our manuscript, we have devised a plan to perform additional histological experiments of *Bdnf* and TrkB expression. Specifically, we will replace the phospho-TrkB antibody with an anti-TrkB antibody to quantify *Bdnf*/TrkB co-expression. Moreover, we acknowledge the concern raised by the reviewers regarding the clarity of some explanations and the potential influence of alternative mechanisms influencing the defects observed in *Bdnf* neurons. We aim to provide a clearer explanation and discussion. We also intend to provide a more comprehensive discussion of the limitations of our LM22A-4 drug treatment experiment. By addressing these points, we wish to ensure that our research is informative to the eLife readership.